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Laser instabilities: a modern perspective

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1. Introduction

As recognized almost immediately following the design of the first laser [1, 2], lasers are dynamical systems capable of exhibiting a wide variety of nonlinear dynamics. For many applications, it is desirable to have a stable, narrow-linewidth, high-intensity, diffraction-limited, light source, but one needs a stable pulse generator in a growing number of applications. Lasers can be used for both applications, but as the output power or the pulse-repetition rate increases, fundamental issues related to the nature of light-matter interaction within the laser become critically important. From a modern perspective, lasers are viewed more and more as interesting nonlinear systems, that can be used as a testing ground for verifying the theory behind the nonlinear dynamical systems. Indeed, when writing a review article on laser dynamics, one is immediately confronted with two different perspectives that researchers in academia and industry have when dealing with laser dynamics. Some years ago, the word 'instability' had a negative connotation in applied research, while the academic researchers considered it an important branch of laser physics. Nowadays, both applied and basic researchers are studying the techniques for controlling laser instabilities, and the control of chaos has become an important topic. Of course, a thorough understanding of laser instabilities is necessary for developing control techniques.

The appearance of spontaneous pulsations, instead of continuous-wave (CW) emission, was observed in a continuously pumped maser by Makhov et al. [3] as early as 1958. Since then, many different types of lasers have been designed and

produced, and by now the use of lasers has become widespread in many scientific disciplines. Even more remarkable is the laser's impact on the society outside science, as witnessed by the popularity of laser-based devices such as CD players, CD-ROM drives in computers, and laser printers. Over the last decade, a global communication revolution has occurred that allows millions of people to be routinely 'on line' on the worldwide network of computers known as the Internet. It is just a matter of time before most communication links become all-optical in nature and use optical signals to transmit digitized data. The two laser systems, without which this revolution cannot take place, are the semiconductor and fiber lasers. A major goal of this review is to describe new kinds of instabilities that can occur in these types of lasers.

Rather than repeating the well-documented historical development of the field of laser instabilities, we refer to several books and review papers. The book chapter by Abraham et al. [4] gives an excellent overview of the dynamical instabilities in lasers, covering the period until 1988. Key topics covered in that review include the semiclassical laser theory for single mode lasers, effects of inhomogeneous broadening, and lasers with saturable absorbers. The 1991 book by Weiss and Vilaseca [5] takes a slightly different approach, by first introducing the basic concepts from the field of nonlinear dynamics. This book also covers three-level gas lasers, multimode lasers, and transverse effects leading to spatial instabilities.

In this review article, we provide an overview of laser instabilities from a modern perspective, in the sense that we focus heavily on the dynamical behavior of semiconductor and fiber lasers, both of which were essential for the development of the lightwave technology during the decade of the 1990s. However, before we deal with such lasers, we discuss in Section 2 the 'quintessential' model of laser dynamics, the so-called Lorenz–Haken model. This model allows us to introduce the notation and the basic concepts behind nonlinear dynamics, while reviewing previous work on laser instabilities in gas lasers. Semiconductor-laser dynamics is addressed in Section 3. Starting with semiconductor Bloch equations, we show how semiconductor lasers can be modeled through a generalized Lorenz–Haken model. We then introduce the rate-equation model for semiconductor lasers, and focus on instabilities that occur when the laser is subjected to optical injection from another laser, or to optical feedback through external reflection. In Section 4, we consider issues that become relevant when dealing with fiber lasers and amplifiers. The most important issue concerns evolution of the intracavity laser field during a single round trip. We discuss such evolution through a Ginzburg–Landau equation, first for a fiber amplifier, and then for fiber lasers after imposing the appropriate boundary conditions. We show how modulation instability evolves from being convective to absolute, because of the feedback occurring in a laser cavity. We then derive a set of multimode rate equations for fiber lasers that include the effects of both the fiber dispersion and nonlinearity and apply them to a simple single-mode case to study the effects of self-phase modulation and intensity-dependent absorption on the Lorenz–Haken-type, self-pulsing instability.

2. Lorenz–Haken model and its assumptions

In principle, a complete description of laser dynamics requires a quantum-mechanical treatment of both the electromagnetic field and the gain medium. For a vast majority of lasers such an approach is not really necessary, and the electromagnetic field can be treated classically because of the large optical intensities involved. Such a 'semiclassical' approach has proven to be quite successful in describing and predicting the dynamical behavior of most lasers. An exception occurs in the case of microcavity lasers, for which the optical field may have to be quantized [6, 7].

In this section, we consider the most simple laser imaginable: a gain medium composed of homogeneously broadened two-level atoms and placed inside a unidirectional ring cavity (see Fig. 1). These simple ingredients turn out to be very powerful in elucidating the basic concepts of laser dynamics, such as the first and the second laser threshold, self-pulsing, Hopf bifurcation, and the development of optical chaos. We begin by deriving the Maxwell–Bloch equations for the simple laser of Fig. 1, and then discuss the Lorenz–Haken model and the instabilities associated with it.

2.1. Maxwell–Bloch equations

The starting point is a collection of identical, non-interacting, two-level atoms, which form the gain medium of our 'quintessential' laser. No real atom can be regarded as a true two-level system, but this simplification is widely used and yields quite reasonable results. We assume that a pumping mechanism exists that maintains a certain population of atoms in the excited state, even in the absence of optical fields.

To avoid complications related to standing waves formed through interference between the backward and forward running waves in a Fabry–Perot cavity, we consider a unidirectional ring-cavity laser (see Fig. 1). A ring cavity, in general, supports both backward and forward running waves (clockwise and counterclockwise), but an optical isolator can be used to suppress one of these

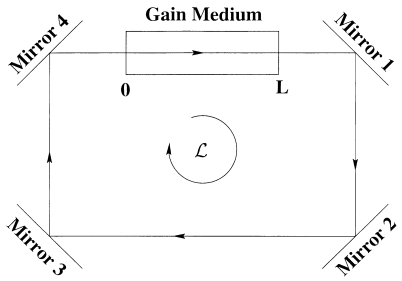


Fig. 1. Schematic of ring-cavity laser of length $\mathcal{L}$ containing a gain medium of length L. One of the mirrors is partially transmitting through which the laser emits light.

waves, making the cavity unidirectional. This choice fixes the longitudinal modes of the cavity, but transverse modes remain unspecified. In practice, most lasers are designed such that one can focus on the fundamental transverse mode and consider one polarization state for the intracavity optical field [4]. Both of these assumptions may not apply to some specific lasers, resulting in spatial and/or polarization instabilities. This review is not intended to cover such laser instabilities, and we make only a few remarks about them in Sections 3 and 4.

2.1.1. Slowly-varying-envelope approximation

Maxwell's equations can be used to derive the wave equation

$$\nabla^2\varepsilon - \frac{1}{c^2}\frac{\partial^2\varepsilon}{\partial t^2} = \mu_0\frac{\partial^2\mathbf{P}}{\partial t^2}, \tag{1}$$

where ε is the electric field vector, $\mathbf{P}$ is the material polarization vector, c is the speed of light in vacuum, and μ_0 is the vacuum permeability. For a plane-polarized traveling wave that maintains its polarization direction throughout the cavity, one can disregard the vector nature of ε and $\mathbf{P}$. The derivatives with respect to the transverse coordinates x and y determine the single-transverse-mode in which the laser operates, but need not to be retained if we assume that the laser mode is not affected by dynamic instabilities. Their neglect results in a much simpler wave equation, that can be written as:

$$\frac{\partial^2\varepsilon}{\partial z^2} - \frac{n_0^2}{c^2}\frac{\partial^2\varepsilon}{\partial t^2} = \mu_0\frac{\partial^2\mathbf{P}}{\partial t^2}, \tag{2}$$

where n_0 is the background refractive index of the host medium in which two-level atoms are assumed to be embedded. Its inclusion is essential for solid-state lasers, but n_0 can be set to 1 for gas lasers. Since the background refractive index n_0 accounts for the host polarization, the quantity $\mathbf{P}$ in Eq. (2) represents the polarization induced by the two-level atoms.

It is useful to express the electric field and the material polarization in the form

$$\varepsilon(z,t) = \frac{1}{2}A(z,t)\exp[i(\beta_0 z - \omega_0 t)] + \text{c.c.}, \tag{3}$$

$$\mathbf{P}(z,t) = \frac{1}{2}B(z,t)\exp[i(\beta_0 z - \omega_0 t)] + \text{c.c.}, \tag{4}$$

where A and B are slowly varying (in both z and t) complex amplitudes of the electric field and the material polarization, respectively, ω_0 is a carrier frequency, $\beta_0 = n_0\omega_0/c$ is the accompanying wavenumber and c.c. stands for complex conjugate. For lasers, ω_0 is often chosen to correspond to the longitudinal mode of the cavity that is closest to the gain peak. By assuming that A and B vary slowly compared with ω_0^{-1} and β_0^{-1}, the wave Eq. (2) for the slowly varying

amplitudes reduces to:

$$\frac{\partial A}{\partial z}+\frac{1}{v_0}\frac{\partial A}{\partial t}=\frac{i\beta_0}{2\varepsilon_0 n_0^2}B, \tag{5}$$

where $v_0 = c/n_0$ is the speed of light in the host medium, assumed to be nondispersive. We discuss in Section 4 how the chromatic dispersion of the host medium can be included in this analysis. Note that we have neglected not only the second-order derivatives of A with respect to t and z, but also the first- and second-order derivatives of B with respect to t. Their neglect constitutes the slowly-varying-envelope approximation (SVEA), which is one of the key approximations in the semiclassical laser theory. Its use is justified for studying laser instabilities as long as the time scale of instabilities is much longer than ω_0^{-1}; this is generally the case in practice.

2.1.2. Rotating-wave approximation

As mentioned earlier, we need to use quantum mechanics when dealing with the dynamics of two-level atoms [8]. Within the electric-dipole approximation (which assumes that the wavelength of the optical field is larger than the dipole size), the Hamiltonian of the two-level system in the presence of the optical field can be written as:

$$\mathcal{H}=\mathcal{H}_0-\mu\cdot\varepsilon, \tag{6}$$

where $\mu = e\mathbf{r}$ is the dipole-moment operator (e is the elementary charge) and $\mathcal{H}_0$ is the unperturbed Hamiltonian of the two-level system with the matrix elements:

$$\langle\varphi_j\mid\mathcal{H}_0\mid\varphi_k\rangle=\hbar\omega_j\delta_{jk}. \tag{7}$$

Here, φ_j ($j = 1, 2$) is an eigenstate of the two-level atom with energy $\hbar\omega_j$. The atomic transition frequency is given by $\omega_A\equiv\omega_2-\omega_1$.

In the presence of the optical field ε, the quantum state $|\psi\rangle$ of the two-level system becomes time dependent and can be written as:

$$|\psi(t)\rangle=c_1(t)\exp(-i\omega_1 t)\mid\varphi_1\rangle+c_2(t)\exp(-i\omega_2 t)\mid\varphi_2\rangle. \tag{8}$$

By using the Schrödinger equation, the coefficients c_1 and c_2 are found to satisfy the following set of two equations:

$$\frac{\mathrm{d}c_1}{\mathrm{d}t}=\frac{ic_2}{\hbar}\exp(-i\omega_A t)\langle\varphi_1\mid\mu\cdot\varepsilon\mid\varphi_{\mathbf{2}}\rangle, \tag{9}$$

$$\frac{\mathrm{d}c_2}{\mathrm{d}t}=\frac{ic_1}{\hbar}\exp(i\omega_A t)\langle\varphi_2\mid\mu\cdot\varepsilon\mid\varphi_{\mathbf{1}}\rangle. \tag{10}$$

Eqs. (9) and (10) are mathematically similar to those appearing in the theory of magnetic resonances. Since that problem was first studied by Bloch [9], the equations that we derive from Eqs. (9) and (10) are known as the optical Bloch equations.

The material polarization can be calculated by using

$$\mathbf{P} = N_{\mathrm{A}}\langle\psi \mid \mu \mid \psi\rangle, \tag{11}$$

where N_{A} is the atomic density. By using Eq. (8), $\mathcal{P}$ becomes

$$\mathbf{P} = N_{\mathrm{A}}[p(t)\mu_{\mathbf{12}} + \mathbf{p}^{*}(\mathbf{t})\mu_{\mathbf{21}}], \tag{12}$$

where

$$p(t) = c_1^{*}(t)c_2(t)\exp(-i\omega_{\mathrm{A}}t), \tag{13}$$

$$\mu_{\mathbf{jk}} = \langle\varphi_{\mathbf{j}} \mid \mu \mid \varphi_{\mathbf{k}}\rangle. \tag{14}$$

Using Eqs. (9) and (10), the microscopic dipole moment, p ,and the inversion probability of a two-level atom defined as $w = |c_2(t)|^2 - |c_1(t)|^2$ satisfy the Bloch equations

$$\frac{\mathrm{d}p}{\mathrm{d}t} = -i\omega_{\mathrm{A}}p + \frac{i}{\hbar}\varepsilon \cdot \mu_{\mathbf{21}}\mathbf{w}, \tag{15}$$

$$\frac{\mathrm{d}w}{\mathrm{d}t} = \frac{2}{i\hbar}\varepsilon \cdot (\mathbf{p}^{*}\mu_{\mathbf{21}} - \mathbf{p}\mu_{\mathbf{12}}). \tag{16}$$

We can write the macroscopic version of these equations by introducing the slowly varying polarization B from Eq. (4) and the population-inversion density $W = N_{\mathrm{A}}w$. The resulting equations are:

$$\frac{\mathrm{d}B}{\mathrm{d}t} = -i(\omega_{\mathrm{A}} - \omega_0)B + \frac{i\mu^2}{2\hbar}W\{A + A^{*}\exp[-2i(\beta_0 z - \omega_0 t)]\}, \tag{17}$$

$$\frac{\mathrm{d}W}{\mathrm{d}t} = \frac{1}{i\hbar}\{AB^{*} - AB\exp[2i(\beta_0 z - \omega_0 t)] - \mathrm{c.c.}\}, \tag{18}$$

where $\mu \equiv |\mu_{12}|$. In view of the SVEA, it is appropriate to neglect in Eqs. (17) and (18) the rapidly oscillating terms at the frequency $2\omega_0$. Their neglect is called the rotating-wave approximation (RWA) and constitutes the second major approximation within the semiclassical laser theory [8].

We complete the description of atomic dynamics by adding a pump term to Eq. (18) and the relaxation terms to Eqs. (5), (17) and (18). It is common to introduce three different relaxation times [8], the population-decay time T_1, the dipole-dephasing time T_2, and the cavity-decay time (also called the photon lifetime) T_{ph}. Several different processes, such as spontaneous emission and atomic collisions, contribute to T_1 and T_2 whereas T_{ph} originates from cavity losses. With these additions, the Maxwell–Bloch equations take their final form:

$$\frac{\partial A}{\partial z} + \frac{1}{v_0}\frac{\partial A}{\partial t} = \frac{i\beta_0}{2\varepsilon_0 n_0^2}B - \frac{A}{2T_{\mathrm{ph}}v_0}, \tag{19}$$

$$\frac{\mathrm{d}B}{\mathrm{d}t} = -\frac{B}{T_2} - i(\omega_\mathrm{A} - \omega_0)B - \frac{i\mu^2}{2\hbar} AW, \tag{20}$$

$$\frac{\mathrm{d}W}{\mathrm{d}t} = \frac{W_0 - W}{T_1} + \frac{1}{\hbar}(AB^* - A^*B), \tag{21}$$

where W_0 is the inversion level that is maintained in the absence of an optical field.

The dynamics of a unidirectional ring-cavity laser is governed by Eqs. (19)–(21), solved with the appropriate boundary condition that accounts for the feedback at cavity mirrors. For a ring cavity of length $\mathcal{L}$ containing a gain medium of length L (see Fig. 1), the boundary condition can be written as:

$$A(0, t) = \sqrt{R_\mathrm{m}} A(L, t - (\mathcal{L} - L)/v_0)\exp(i\phi_{RT}), \tag{22}$$

where R_m is the reflectivity of the output mirror, and all other mirrors in Fig. 1 are assumed to be 100% reflective. The longitudinal modes of the cavity are determined by requiring that the round-trip phase shift ϕ_{RT} be an integer multiple of 2π.

For the discussion of laser instabilities, it is useful to rescale the optical field A, the medium polarization B, and the population inversion W as:

$$\hat{A} = \sqrt{\frac{\varepsilon_0 c n_0}{2}} A, \tag{23}$$

$$\hat{B} = \frac{\beta_0}{\varepsilon_0 n_0^2} \sqrt{\frac{\varepsilon_0 c n_0}{2}} B, \tag{24}$$

$$g = \sigma_\mathrm{s} W, \tag{25}$$

where the transition cross section, σ_s is defined as:

$$\sigma_\mathrm{s} = \frac{\mu^2 \omega_0 T_2}{2\varepsilon_0 \hbar c n_0}. \tag{26}$$

With this rescaling, $|\hat{A}|^2$ has units of intensity ($\mathrm{W/m^2}$), $\hat{B}$ has units of $\sqrt{\mathrm{W}}/\mathrm{m}^2$, and g represents the power gain per unit length. After omitting the hats on $\hat{A}$ and $\hat{B}$ for notational simplicity, the rescaled equations for the field, polarization and gain become:

$$\frac{\partial A}{\partial z} + \frac{1}{v_0}\frac{\partial A}{\partial t} = \frac{i}{2} B - \frac{A}{2T_\mathrm{ph} v_0}, \tag{27}$$

$$T_2 \frac{\mathrm{d}B}{\mathrm{d}t} = -(1 - i\delta)B - iAg, \tag{28}$$

$$T_1 \frac{\mathrm{d}g}{\mathrm{d}t} = g_0 - g + \frac{\mathrm{Im}(A^* B)}{I_{\mathrm{sat}}}, \tag{29}$$

where $g_0 = \sigma_s W_0$ is the small-signal gain, and the scaled atomic detuning δ and the saturation intensity I_{sat} are defined as:

$$\delta = (\omega_0 - \omega_A) T_2, \quad I_{\mathrm{sat}} = \frac{\hbar^2 c n_0 \varepsilon_0}{2\mu^2 T_1 T_2}. \tag{30}$$

2.2. Lorenz–Haken equations

The steady-state solutions of Eqs. (27)–(29) satisfying the boundary condition in Eq. (22) can be found rather easily [10–12]. A stability analysis of these steady-state solutions has proved to be a hard nut to crack, but Lugiato et al. [13] succeeded in 1986 in making some progress. The main problem is that Eq. (27) allows for longitudinal variations of the intracavity optical field along the cavity length, but their inclusion complicates severely a stability analysis of the steady-state solutions. An alternative approach makes use of a modal decomposition, by noting that the intracavity laser field is composed of one or more longitudinal modes of the cavity. Many modern laser systems, such as fiber lasers, have a long cavity length ($\backsim$10 m), together with a large gain bandwidth ($\backsim$1 THz), allowing excitation of thousands of longitudinal modes simultaneously. We discuss in Section 4 to what extent the issue of stability can be addressed analytically in such lasers. Here, we focus on lasers operating in a single, or at most, a few longitudinal modes.

2.2.1. Mean-field limit

Most linear-stability studies avoid dealing with the $\partial A/\partial z$ term in Eq. (27) and employ what has become known as the mean-field limit. By taking this limit, the field Eq. (27) is split into a set of purely temporal equations, each of which governs the evolution of a single longitudinal mode. Although limited in its scope, the mean-field approximation has proved to be quite successful in studying the stability of lasers. However, one should be aware that its predictions for any real laser need to be checked against numerical solutions of the exact Maxwell–Bloch equations.

It is easy to see from Eq. (27) that longitudinal intensity variations in the steady state diminish as cavity losses per round trip decrease. Cavity losses can be reduced by increasing the mirror reflectivity R_m, such that the ratio $T_{ph} v_0 (1 - R_m)/\mathcal{L}$ remains finite [14]. A 'uniform intensity' does not guarantee a 'uniform field', because longitudinal variations of the phase can be substantial even if the intensity is virtually constant. However, unless there are intracavity elements that provide large phase changes, the optical phase is expected not to vary much over a single round trip [the large phase shift $\beta_0 \mathcal{L}$ has been factored out through Eq. (3)]. Since mirror losses are assumed to be relatively small, the

boundary-value problem simplifies considerably if the *localized* mirror losses can be distributed along the cavity length, so that the total cavity loss becomes

$$\alpha = \frac{1}{2T_{\rm ph}v_0} = \alpha_{\rm int} + \frac{1}{\mathcal{L}} \ln\left(\frac{1}{R_{\rm m}}\right), \tag{31}$$

where $\alpha_{\rm int}$ is the internal loss of the laser cavity and has been added phenomenologically to account for other sources of intracavity losses.

The Lorenz–Haken model corresponds to a laser forced to oscillate in a single longitudinal mode (as well as in a single transverse mode), and whose optical gain is provided by a homogeneously broadened two-level atomic system. This model can be obtained from Eqs. (27)–(29) by taking the mean-field limit. Using $\bar{A}$ and $\bar{B}$ to denote the mean field and mean polarization, respectively, and setting $\partial\bar{A}/\partial z = 0$, since $\bar{A}$ is z independent by definition, the resulting equations become

$$\frac{{\rm d}\bar{A}}{{\rm d}t} = \frac{i}{2} v_0 \bar{B} - \frac{\bar{A}}{2T_{\rm ph}}, \tag{32}$$

$$T_2 \frac{{\rm d}\bar{B}}{{\rm d}t} = -(1 - i\delta)\bar{B} - i\bar{A}g, \tag{33}$$

$$T_1 \frac{{\rm d}g}{{\rm d}t} = g_0 - g + \frac{{\rm Im}(\bar{A}^*\bar{B})}{I_{\rm sat}}. \tag{34}$$

It was shown by Haken [15] in 1975 that Eqs. (32)–(34) are isomorphic to the Lorenz equations [16] describing convective fluid flow (in the Rayleigh–Bénard configuration). Almost ignored in 1963, Lorenz equations became the pole bearer for the emerging field of nonlinear dynamics. Although laser instabilities were considered even before 1963, and several studies [17–19] predicted numerically an intrinsic instability of laser models, such predictions were mostly ignored by the laser community. It was only after Haken established in 1975 the mathematical isomorphism between the hydrodynamic Lorenz equations and the optical Maxwell–Bloch equations that the words 'optical chaos' became common in the laser literature.

Eqs. (32)–(34) can be written in a form that is identical to the Lorenz equations by introducing $t' = t/T_2$ as a normalized time. The resulting equations, called the Lorenz–Haken equations, take the form:

$$\frac{{\rm d}x}{{\rm d}t} = -\sigma(x - y), \tag{35}$$

$$\frac{{\rm d}y}{{\rm d}t} = -(1 - i\delta)y + (r - z)x, \tag{36}$$

$$\frac{{\rm d}z}{{\rm d}t} = -bz + {\rm Re}(x^*y), \tag{37}$$

where the prime over t' has been dropped for notational simplicity. The two parameters σ and b are the ratios of the damping rates and are defined through:

$$\sigma = T_2/2T_{\mathrm{ph}}, \quad b = T_2/T_1. \tag{38}$$

The other two parameters of the problem are the pump term r and the detuning parameter δ, defined as:

$$r = g_0 v_0 T_{\mathrm{ph}}, \quad \delta = (\omega_0 - \omega_{\mathrm{A}})T_2. \tag{39}$$

The normalized variables x, y and z represent the optical field, the induced polarization, and the optical gain, respectively, and are defined as:

$$x = (b/I_{\mathrm{sat}})^{1/2}\bar{A}, \tag{40}$$

$$y = (b/I_{\mathrm{sat}})^{1/2}(iv_0 T_{\mathrm{ph}})\bar{B}, \tag{41}$$

$$z = v_0 T_{\mathrm{ph}}(g_0 - g). \tag{42}$$

The Lorenz–Haken Eqs. (35)–(37) thus involve four real parameters (σ, b, r, δ) and three dynamic variables x, y and z, two of which can become complex. These equations are simple enough to make some analytical progress, yet contain sufficient physics that they can predict a wealth of nonlinear dynamical effects.

2.2.2. First laser threshold

Every study of nonlinear dynamics typically begins with an investigation of the 'fixed points' in the phase space, which represent the steady-state solutions, and their stability to small perturbations. The fixed points for the Lorenz–Haken equations correspond to the continuous-wave (CW) operation of a laser. Note that the CW operation with a constant power does not represent a steady state in a strict sense, because the optical phase will vary linearly with time if the mode frequency changes from its empty-cavity value of ω_0. However, one can always rescale the dynamical equations such that each CW solution becomes a true fixed point.

Eqs. (35)–(37) have a trivial steady-state solution:

$$x_{\mathrm{s}} = y_{\mathrm{s}} = z_{\mathrm{s}} = 0. \tag{43}$$

This solution may seem uninteresting since no light is emitted by the laser (except for spontaneous emission, not included in the Lorenz–Haken model). However, its stability properties readily provide us with the value of the pump parameter at which the lasing starts. This value is called the laser threshold, but the term 'first threshold' is often used in the literature on laser instabilities.

The stability of the trivial solution is analyzed by using a standard technique known as the *linear stability analysis*. After perturbing the fixed point of Eq. (43) so that $x(t) = x_{\mathrm{s}} + \delta x(t)$, $y(t) = y_{\mathrm{s}} + \delta y(t)$, and $z(t) = z_{\mathrm{s}} + \delta z(t)$, we obtain the following evolution equations for the perturbations δx, δy and δz:

$$\delta\dot{x} = -\sigma(\delta x - \delta y), \tag{44}$$

$$\delta\dot{y} = -(1 - i\delta)\delta y - (r - \delta z)\delta x, \tag{45}$$

$$\delta\dot{z} = -b\delta z + \mathrm{Re}(\delta x^{*}\delta y), \tag{46}$$

If the initial perturbation is assumed to be very small, we can neglect the nonlinear terms $\delta z\delta x$ in Eq. (45) and $\delta x^{*}\delta y$ in Eq. (46). This process is called 'linearization' as it leads to a set of linear equations, which can be solved using the Laplace-transform technique. In this technique, each perturbation is assumed to evolve exponentially with time as $\exp(st)$, where s is the (complex) Laplace variable. A nontrivial solution of the resulting set of algebraic equations exists only if the determinant of the coefficient matrix vanishes, i.e.

$$\begin{vmatrix} s+\sigma & -\sigma & 0 \\ -r & s+1-i\delta & 0 \\ 0 & 0 & s+b \end{vmatrix} = 0. \tag{47}$$

When any solution of Eq. (47) has a positive real part, the steady state is unstable against small perturbations, since they grow exponentially with time. In fact, the real part of the solution s governs the growth rate of perturbation, while the imaginary part provides the frequency when the growth of perturbation follows an oscillatory pattern.

The characteristic equation obtained by expanding the determinant in Eq. (47) has three roots, one of which $s = -b$ is real and negative, and can thus be ignored. The other two roots are solutions of a quadratic equation:

$$s^2 + (\sigma + 1 - i\delta)s - \sigma(r - 1 + i\delta) = 0, \tag{48}$$

and are complex conjugate of each other. The pump value r at which the nonlasing state loses its stability (i.e. the first threshold) can be found by substituting $s = i\Omega$ in Eq. (48). Since the growth rate of perturbation is then zero, we look for the pump value at which the nonlasing state is marginally stable such that an indefinitely small increase in the pump r will destabilize the steady state. This value is found to be

$$r_{\mathrm{th}}^{(1)} = 1 + \frac{\delta^2}{(\sigma + 1)^2}. \tag{49}$$

In the absence of detuning ($\delta = 0$), $r_{\mathrm{th}}^{(1)} = 1$ or $g_0 = (v_0 T_{\mathrm{ph}})^{-1} = \alpha_{\mathrm{c}}$ from Eq. (39). This is a well-known result from laser theory stating that the gain must equal the cavity loss before lasing can begin. When the atomic system is detuned from the cavity resonance, the first laser threshold increases, simply because the gain is reduced at the laser frequency and more pump energy is needed to compensate for cavity losses.

2.2.3. Second laser threshold

Beyond the first threshold, the laser begins to emit CW power. The amount of power is found from the non-trivial steady-state solution of Eqs. (35)–(37). However, we should allow for a frequency shift of the laser mode by the gain medium, a phenomenon referred to as mode pulling in the laser literature. Thus, we look for a CW solution of the form:

$$x_s = X_0 \exp[-i(\Delta\omega_s t + \varphi_s)], \tag{50}$$

$$y_s = Y_0 \exp(-i\Delta\omega_s t), \tag{51}$$

$$z_s = Z_0, \tag{52}$$

where $\Delta\omega_s$ is a possible frequency shift of the laser mode from the cavity-resonance frequency ω_0, and φ_s is the phase lag between the polarization and the electric field. By substituting Eqs. (50)–(52) in the Lorenz–Haken equations, we obtain the following CW solution:

$$\Delta\omega_s = -\delta\sigma/(\sigma + 1), \tag{53}$$

$$\tan\varphi_s = \delta/(\sigma + 1), \tag{54}$$

$$Z_0 = r - r_{th}^{(1)}, \tag{55}$$

$$X_0 = \pm\sqrt{bZ_0}, \tag{56}$$

$$Y_0 = \pm\sqrt{r_{th}^{(1)} bZ_0}. \tag{57}$$

From Eqs. (56) and (57), the (normalized) laser power becomes $X_0^2 = b(r - r_{th}^{(1)})$, indicating that the output power increases linearly with pumping after the first laser threshold, again a well-known result in the laser theory. The important question is whether the CW state given by Eqs. (53)–(57) is stable for all $r > r_{th}^{(1)}$. In some regions of the parameter space, the answer is indeed no, and that is where interesting nonlinear dynamics, such as self-pulsing and chaos, can occur.

We therefore perform a linear stability analysis of the CW solution given by Eqs. (53)–(57). Following the linear-stability technique outlined earlier, we obtain the following cubic polynomial equation for the resonant case ($\delta = 0$):

$$s^3 + c_2 s^2 + c_1 s + c_0 = 0, \tag{58}$$

where the coefficients are given by

$$c_2 = \sigma + b + 1, \tag{59}$$

$$c_1 = b(\sigma + r), \tag{60}$$

$$c_0 = 2b\sigma(r-1). \tag{61}$$

The critical pump value at which the CW solution becomes unstable is found by looking for roots of the form $s = i\Omega$. Substituting $s = i\Omega$ in Eq. (58), we obtain the following expression for the critical pump value:

$$r_{\text{th}}^{(2)} = \frac{\sigma(\sigma+b+3)}{\sigma-b-1}. \tag{62}$$

This pump value $r_{\text{th}}^{(2)}$ is called the *instability threshold* or the *second threshold* of the laser. When the pump r exceeds this threshold, the CW state becomes unstable through a Hopf bifurcation, and the laser begins to emit pulses even when pumped at a constant rate r. The frequency of self-pulsing, or the repetition rate of pulses emitted by the laser, is given by:

$$\Omega = \sqrt{b(\sigma + r_{\text{th}}^{(2)})}. \tag{63}$$

Since the linear-stability analysis can only predict the initial dynamics close to the second threshold, numerical simulations of the full Lorenz–Haken equations should be carried out to find the shape of emitted pulses and to study how the self-pulsing state evolves with a further increase in pumping.

Before we address this issue, we note that a second threshold does not exist for all combinations of the parameters σ and b. From Eq. (62) it is obvious that the second threshold is usually quite high. Its minimum value, reached for $\sigma = 3$ and $b = 0$, is 9, i.e. the laser should be pumped at least nine times above the first threshold for self-pulsing to occur. Since $r_{\text{th}}^{(2)}$ is positive by definition, Eq. (62) provides us with a necessary condition for the second threshold to exist:

$$\sigma > b + 1. \tag{64}$$

In terms of the photon lifetime T_{ph} and the atomic lifetimes T_1 and T_2, this condition becomes:

$$(2T_{\text{ph}})^{-1} > T_1^{-1} + T_2^{-1}. \tag{65}$$

The condition In Eq. (64) is known as the 'bad-cavity condition' since it requires a rather lossy cavity for the second laser threshold to exist.

The bad-cavity condition is usually regarded as a necessary condition for laser instabilities to occur. One should keep in mind, however, that this is only true in a linear-stability context that requires small perturbations. It was shown by Narducci et al. [20] and Casperson [21] that there is a clear distinction between small and large perturbation of the stationary state. In the case of a large perturbation, also known as 'hard-mode excitation', the laser can exhibit instabilities for pump values below $r_{\text{th}}^{(2)}$. However, when the condition in Eq. (64) is satisfied, one can be sure the laser will exhibit instabilities when $r > r_{\text{th}}^{(2)}$.

If the laser is detuned from the atomic resonance, i.e. when $\delta \neq 0$, the analysis becomes much more complex [22, 23]. Zeghlache and Mandel [22] analyzed the detuned case and derived a power series in the detuning δ for the second

threshold, resulting in the following expression for the case of $\sigma = 3$ and $b = 1$:

$$r_{\text{th}}^{(2)}(\delta) = 3 + 18(1 + 0.6\delta^2 + 0.81\delta^4)^{1/2} + 16.8\delta^2. \tag{66}$$

It reduces to $r_{\text{th}}^{(2)} = 21$ for $\delta = 0$ in agreement with Eq. (62).

At $r = r_{\text{th}}^{(2)}$ the system undergoes a Hopf bifurcation that is subcritical for small detunings ($\delta^2 \leqslant 1/3$) and supercritical for large detunings ($\delta^2 > 1/3$). Supercritical means that the nonlinear and linear terms in the 'normal-form' description of the problem have opposite signs [5]. The resonant Lorenz–Haken equations show a subcritical Hopf-bifurcation at $r = r_{\text{th}}^{(2)}$. With a suitable transformation, the detuned case can be described by the resonant Lorenz–Haken equations by using a complex pump parameter r_{eff} whose imaginary part depends on the amount of detuning [23]. Exact analytical expressions for the second threshold of a detuned laser can be derived by this approach.

2.2.4. Nonlinear dynamics beyond the second threshold

Owing to the fame of the Lorenz–Haken equations, a large number of publications have appeared, describing the dynamical peculiarities of these equations. We do not intend to give a complete review of these results, but instead use the Lorenz–Haken equations to introduce some basic terms and phenomena that will be needed in later sections. We restrict ourselves to the resonant Lorenz–Haken equations, i.e. we set $\delta = 0$. We choose the parameter values $\sigma = 1.4253$ and $b = 0.2778$, which correspond to a far-infrared NH_3 laser [24, 25]. From Eq. (62), such a laser has a second threshold at $r_{\text{th}}^{(2)} = 45.446$. Although it has been argued that the atomic transitions responsible for the lasing action in NH_3 lasers are not as simple as those occurring for a two-level system, and the laser dynamics should be described by a set of nine rather than three equations, the experimental results are quite similar to the predictions of the Lorenz–Haken equations [5].

Starting at $r = 0$, different types of nonlinear dynamics are encountered with an increase in the pump level. In the regime $0 < r < r_{\text{th}}^{(1)} = 1$, only one fixed point exists, the nonlasing state of Eq. (43), which is stable, and the laser remains 'off'. In reality, because of the ever-present spontaneous emission, the laser emits some light through luminescence. This is incoherent light and is not included in the Lorenz–Haken equations, since spontaneous emission has been neglected. However, it can be easily incorporated through the addition of the Langevin-noise terms [26]. Since these terms are stochastic in nature, it is common to omit them to remain focused on the deterministic aspects of the nonlinear laser dynamics.

Beyond the first laser threshold ($r > r_{\text{th}}^{(1)}$), two CW solutions are possible, while the nonlasing state is unstable. The laser emits constant power and, provided the pump remains smaller than the second threshold $r_{\text{th}}^{(2)}$, stable CW operation is observed. Small perturbations, e.g. those caused by spontaneous emission, damp with time following an oscillatory behavior. The decay rate and the frequency of this relaxation process are given by the real and imaginary parts of the solution s of the characteristic Eq. (58). Since spontaneous emission is constantly exciting

these oscillations, a peak at the relaxation–oscillation frequency is observed in the power spectrum of the stable single-mode laser. Interestingly, the basin of attraction of the two fixed points does not encompass the entire phase space (x, y, z) for all pump values smaller than $r_{\mathrm{th}}^{(2)}$. As the laser is pumped beyond its first threshold, the laser evolves toward one fixed point (and stays there) as long as

$$r_{\mathrm{th}}^{(1)} < r < r_{\mathrm{A}} < r_{\mathrm{th}}^{(2)}, \tag{67}$$

where $r_{\mathrm{A}} = 35.85$ for the parameters chosen. In the range $r_{\mathrm{A}} < r < r_{\mathrm{th}}^{(2)}$ the two fixed points are still stable, but trajectories in the phase space do not end up at either one of them, if started at the nonlasing state of Eq. (43). Instead, the system spirals around one fixed point for some time, then switches to spiraling around the other fixed point, and keeps on switching back and forth. This kind of trajectory, known as a 'strange attractor' is shown in Fig. 2. Still, for pump values smaller than $r_{\mathrm{th}}^{(2)}$ the fixed points are stable, and the system will remain on a fixed point *if it is prepared in that state*. The coexistence of stable fixed points and the strange attractor makes the bifurcation at $r_{\mathrm{th}}^{(2)}$ subcritical.

When the pump exceeds the second laser threshold, the relaxation oscillations grow rather than dampen with time as the two fixed points have lost their stability. Any perturbation, no matter how small, will force the system away from the two fixed points. More precisely, if perturbations caused by spontaneous emission extend beyond the basin of attraction of the fixed points, the laser will

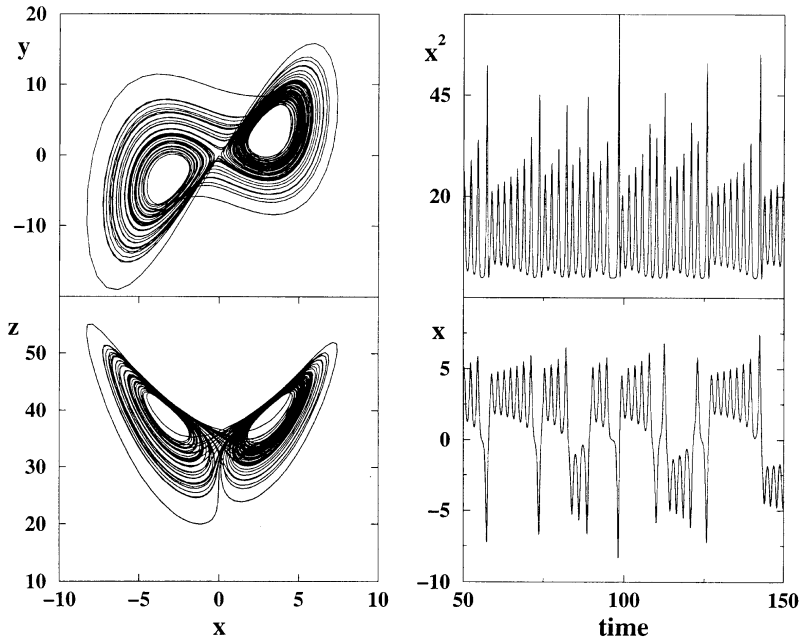


Fig. 2. Phase-space (left column) and temporal (right column) dynamics of the Lorenz–Haken equations with parameters $\sigma = 1.4253$, $b = 0.2778$, $\delta = 0$, and $r = 40$.

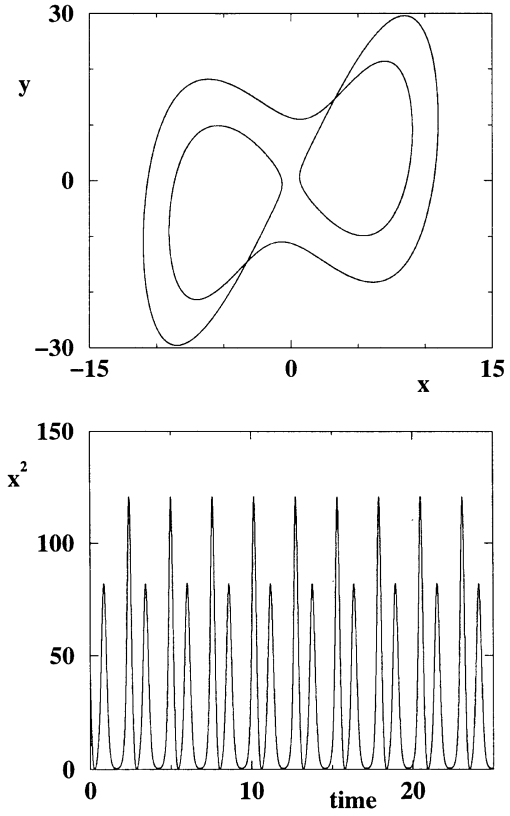


Fig. 3. Period-2 dynamics under the conditions of Fig. 2 except that the pump parameter $r = 112.5$ lies in the second periodic window occurring after the second threshold.

cease to operate CW with a constant power. Depending on how far above the second threshold the laser is pumped, the system can exhibit quite different dynamic behavior. For pump values not too far above the second threshold, the laser shows the strange behavior already discussed for the regime $r_A < r < r_{th}^{(2)}$. However, periodic self-pulsing windows can appear in some range of pumping levels; exact values, of course, depend on the parameters σ and b. In these pump ranges, the trajectory closes upon itself, and the laser intensity shows self-pulsing behavior. Fig. 3 shows as an example of such a self-pulsing behavior for $r = 112.5$. This value of r lies in the second periodic window covering the range $107.3 < r < 117.8$ (the first periodic window is found to occur for $73.4 < r < 73.5$). When the pump exceeds a value of $r > 138$, a fundamental change occurs: the phase-space trajectory loses its strangeness, and eventually the laser ends up in a simple period-1 orbit through a sequence of inverse period-doubling bifurcations (at $r \approx 146.5$). Moreover, in the range $146.5 < r < \infty$ the trajectory remains a simple closed orbit, indicative of the self-pulsing behavior of the laser intensity. It is worthwhile to go through the steps associated with the

sequence of period-doubling bifurcations as r decreases. For pump values slightly smaller than $r = 146.5$, the orbit closes upon itself not after one but two round-trips, and the oscillation period doubles. This behavior is referred to as the period-doubling bifurcation. The next period-doubling bifurcation is found to occur at $r \approx 141.5$. Fig. 4 shows the period-4 dynamics of the field intensity for $r = 140$. Such bifurcations appear with an increasing rate as r decreases further, and at $r \approx 139.6$ the periodicity of the orbit becomes infinite. Finally, fully developed chaos occurs close to $r \approx 138$.

2.2.5. Three routes to chaos

In the example above, a period-doubling route to chaos was found to occur as a control parameter was varied. However, the period-doubling route, also called the Feigenbaum scenario, is not the only way a system can become chaotic. Although the precise number of routes to chaos is not known, three scenarios that appear quite often are known as (i) the period-doubling route, (ii) the quasi-periodic route, and (iii) the intermittency route [5]. It goes beyond the scope of

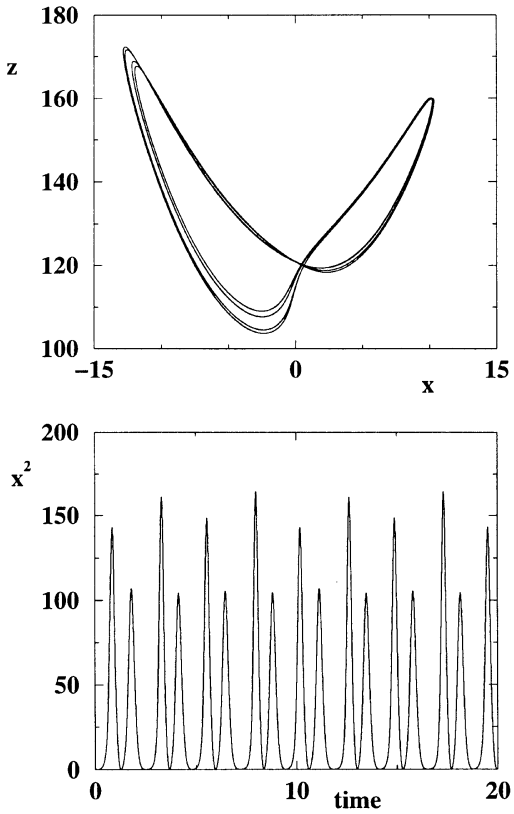


Fig. 4. Period-4 dynamics under the conditions of Fig. 2 except that the pump parameter $r = 140$ lies in the inverse period-doubling bifurcation regime.

this paper to deal with each of them in detail, and only a short description is provided.

The quasi-periodic route to chaos, also known as the Ruelle–Takens–Newhouse scenario, consists of a sequence of three bifurcations occurring at three values of the control parameter ρ. At $\rho = \rho_1$, the attractor in the phase space changes from a fixed point to a periodic orbit, at ρ_2 it changes to a torus because of the presence of two incommensurate frequencies, and at ρ_3 a third independent frequency appears such that the attractor becomes a hypertorus. In many cases, this attractor is unstable against perturbations, and the system dynamics becomes chaotic. The 'intermittency' route to chaos, also known as the Pomeau–Manneville scenario, requires only one bifurcation point, beyond which the self-pulsing oscillations (periodic orbit) appear to be interrupted at random times by turbulent phases. Three types of intermittency are known, classified through the stability analysis of the periodic orbit (the so-called Floquet analysis).

2.3. Lorenz–Haken model and real lasers

Since 1975, many systematic studies have been reported on the Lorenz–Haken dynamics and its generalizations. A valid question is whether real lasers exist that can be modeled by the Lorenz–Haken equations. As discussed earlier, rather severe approximations were necessary to obtain the Lorenz–Haken equations from the Maxwell–Bloch equations. Not only were the SVEA and RWA made, but we also restricted ourselves to a single cavity mode (both transverse and longitudinal) in a unidirectional ring-cavity configuration, and assumed that the gain medium can be modeled by homogeneously broadened two-level atoms.

It is difficult to satisfy all of these assumptions for a real laser. Indeed, more than 20 years after their appearance, only one type of lasers appears to be well modeled by the Lorenz–Haken equations—the far-infrared NH_3 laser [24, 25]. Although the atomic system providing the gain in such lasers is really a *three*-level system, their dynamics is in qualitative and quantitative agreement with the Lorenz–Haken model [5]. Even though most lasers do not satisfy all the assumptions made by the Lorenz–Haken model, what makes this model so successful is the observation that the dynamics exhibited by a large number of lasers shows a strong resemblance with the Lorenz–Haken dynamics [5]. This fact makes the Lorenz–Haken model an excellent and relevant model that allows a relatively simple investigation of the nonlinear dynamics of most lasers.

While searching for other 'Lorenz–Haken' lasers, one is very quickly confronted with the fact that the bad-cavity condition in Eq. (64) is a very demanding one. It turns out that the characteristic lifetimes (T_{ph}, T_1 and T_2) for most laser systems are such that the bad-cavity condition is not satisfied. It is possible to construct intentionally a bad-cavity laser if the gain medium can provide high gain per pass. An example of such a laser is the He–Xe laser. This laser has been successfully used to investigate the fundamental (quantum-mechanical) aspects of the laser linewidth [27].

2.3.1. Classification scheme for lasers

A question of interest is how to classify different lasers from the standpoint of their nonlinear dynamics. Arecchi et al. [28, 29] introduced a classification scheme for homogeneously broadened single-mode lasers based on the relative magnitudes of the three lifetimes. The class-A lasers (e.g. dye lasers) are described by the field Eq. (35), since the atomic lifetimes T_2 and T_1 are so small compared with the photon lifetime T_{ph} that both the polarization and the population inversion [Eqs. (36) and (37)] can be adiabatically eliminated. For class-B lasers (e.g. semiconductor lasers) only the polarization Eq. (36) can be adiabatically eliminated, while for class-C lasers (e.g. far-infrared NH_3 laser) all decay rates are of the same order of magnitude.

A prerequisite for chaotic dynamics to occur is the existence of three coupled, first-order, nonlinear differential equations. As a result, chaotic behavior is ruled out for class-A lasers, whose dynamics are governed by one (complex) nonlinear equation. The class-B lasers, being described by one complex and one real equation, would seem to satisfy the prerequisite for chaotic dynamics. However, when the complex equation for the optical field is written as two real equations (governing the intensity and phase dynamics), it turns out that the phase equation is decoupled from the other two describing optical intensity and population inversion, i.e. the phase is a *slaved* dynamic variable. As a result, class-B lasers are incapable of exhibiting chaos, and only class-C lasers are the candidates for chaotic dynamics. One should realize, however, that this classification scheme is only approximate. As was shown by Mandel et al. [30], a rather extensive set of mathematical conditions has to be satisfied to justify adiabatic elimination of a dynamic variable.

One way to bring the optical chaos back into a class-B laser is to add more degrees of freedom to its dynamics, e.g. through pump modulation, external injection, or optical feedback. Intentional pump modulation is used in many applications, and unintentional optical feedback occurs naturally in many practical systems. For this reason, chaotic dynamics in semiconductor lasers has been a subject of intense investigation for the last two decades [31]. Pump modulation of an erbium-doped fiber-ring laser results in chaotic dynamics through all three routes to chaos, as also observed experimentally [32].

Another way of destabilizing the CW state and producing pulses from a laser, often implemented for generating ultrashort optical pulses, consists of inserting a nonlinear element within the laser cavity (e.g. a saturable absorber). It was realized in 1978 that a laser with saturable absorber can be modeled by using equations that are similar to those used for modeling hydrodynamic Rayleigh–Bénard convection [33]. Since then, such lasers have been investigated in considerable detail [5, 34]. From a practical standpoint, the use of a saturable absorber within a laser cavity can result in a reliable pulse source, and most self-pulsing semiconductor lasers make use of this technique [35]. It should be stressed that saturable absorbers are also often used to induce passive mode-locking [36]. However, this phenomenon cannot be described by the single-mode Lorenz–Haken model as it requires more than one longitudinal mode by definition.

Theoretical models for passive mode-locking generally use a spatio-temporal model by retaining the z-dependent term in Eq. (5) since the mean-field approximation is no longer valid [37].

The difference between the phenomena of self-pulsing and passive mode-locking, both of which produce a regular pulse train, can be understood as follows. Self-pulsing can be described by a single-mode, rate-equation model, and the pulse width is by definition much longer than the cavity round-trip time. The repetition rate of pulses is related to the frequency of undamped relaxation oscillation and is always smaller than the longitudinal-mode spacing. In contrast, mode-locking-induced pulsations have none of these restrictions. The pulse width can be a small fraction of the round-trip time within the laser cavity, and the repetition rate can be a high integer multiple of the longitudinal-mode spacing. A recent example of such harmonic mode-locking is provided by the distributed-Bragg-reflector semiconductor laser containing an intracavity saturable absorber, locking on the 40th harmonic of the round-trip frequency, and producing mode-locked pulses at a 1.54 THz repetition rate [38].

2.3.2. Generalizations of the Lorenz–Haken model

The Lorenz–Haken model has been generalized in several different directions. It is not possible to discuss here all extensions of this model, and we refer to several reviews [4, 39, 40] and books [5, 41–43] that cover this topic. One phenomenon that reduces the second threshold considerably is inhomogeneous broadening, occurring when the two-level atoms that constitute the gain medium have slightly different resonance frequencies. For gas lasers, this spread in resonance frequencies is due to different atomic velocities. Mathematically, the inclusion of inhomogeneous broadening transfers the Lorenz–Haken equations into a set of integro-differential equations, which have many more degrees of freedom compared with the homogeneously broadened case. Casperson showed in 1978, both theoretically and experimentally, that the second threshold of an inhomogeneously broadened, high-gain, He–Xe laser is substantially lower than that of homogeneously broadened lasers [44]. The 'Casperson instability' is generally recognized as the first experimentally observed laser instability that can be understood both qualitatively and quantitatively by the semiclassical laser theory [45, 46].

Several other issues are attracting considerable attention in recent years. Firstly, most lasers employ Fabry–Perot cavities, and an extension of the Lorenz–Haken equations to such cavities is desirable [47, 48]. This extension is not trivial, since the formation of standing waves in Fabry–Perot cavities introduces longitudinal intensity variations on a wavelength scale, that make the application of the mean-field limit questionable. Secondly, fiber lasers have relatively long cavities ($\backsim$10 m) in which the optical field exhibits considerable *intracavity* variations during a single round trip because of fiber dispersion and nonlinearities, making the mean-field limit again questionable. We will return to this issue in Section 4 where we discuss fiber-laser instabilities. Thirdly, the generalization of the Lorenz–Haken

model to semiconductor lasers will be quite useful in view of the widespread use of such lasers in practice. Since semiconductor lasers are a far cry from a homogeneously broadened two-level system, such an extension is far from being obvious. The next section deals with instabilities in semiconductor lasers.

3. Semiconductor lasers

Since the demonstration of semiconductor lasers in 1962 by four groups [49–52], a large variety of semiconductor-based lasers have been developed for commercial and scientific applications. Different semiconductor materials are being used to produce laser emission at wavelengths ranging from the infrared to the near-ultraviolet region of the optical spectrum. Semiconductor lasers are not only used as CW light sources, but they are made to emit short pulses at a stable repetition rate in an increasing number of applications. Some of the recent advances include: (i) the development of GaN-based blue semiconductor lasers [53], (ii) the emission of high CW power (>150 W) from semiconductor-laser arrays [54], and (iii) the demonstration of 1.54 THz mode-locked pulses from a semiconductor laser [54]. Several books have addressed various aspects of semiconductor lasers [55–57]. Here, we focus on two issues relevant to our discussion of laser instabilities.

Firstly, we introduce the semiconductor Bloch equations, which are substantially more complicated than the Bloch equations of Section 2 obtained for a homogeneously broadened two-level system. The characteristic time scale for polarization dynamics is on the femtosecond scale in semiconductor lasers, making the material polarization a suitable candidate for adiabatic elimination. However, many applications nowadays involve ultrashort pulses, and the optical response of semiconductor materials on femtosecond time scales has become a relevant issue. Recently, attempts have been made to 'generalize' the Lorenz–Haken equations such that the basic features of the semiconductor-laser gain dynamics are adequately described without requiring huge computational resources that are needed for full-scale simulations. Such simple models are quite interesting from a nonlinear-dynamics point of view, and we discuss one of them in detail.

Secondly, we consider single-mode semiconductor-laser dynamics on a time scale much longer than 1 ps, and carry out adiabatic elimination of the material polarization. The dynamics of single-mode semiconductor lasers is then well described by a set of two rate equations similar to those obtained for the other class-B laser. As discussed earlier, class-B lasers do not exhibit a second threshold. However, the nonlinear dynamics of such lasers changes considerably if more degrees of freedom are added optically, through injection from an external laser source or through optical feedback. The latter has been a 'hot topic' since the early 1980s, and continues to attract considerable attention. We discuss the instability issues related to both optical injection and optical feedback.

3.1. Semiconductor Bloch equations

The gain in a semiconductor laser is provided by a pump mechanism that involves current injection across a forward-biased p–n junction, leading to a steady-state population of electron–hole pairs inside the 'active' layer. These electron–hole pairs recombine through spontaneous and stimulated emission processes to generate light, similar to the case of an excited two-level system [5].

Semiconductor gain media are complicated objects that cannot be modeled as a simple homogeneously broadened two-level system used in Section 2. Although the optical gain is provided by electron–hole pairs that can be modeled as a two-level system, their different kinetic energies (inhomogeneous broadening) and their interaction with the semiconductor lattice (band structure) and with each other (Coulomb interaction and the associated many-body effects), severely complicate the analysis. As a result, the semiconductor Bloch equations are substantially more complicated than the simple Bloch equations derived in Section 2 for a two level system. Starting with the microscopic many-body theory under the Hartree–Fock approximation, and using the single-plasmon-pole model for plasma screening [58, 59], the following semiconductor Bloch equations are obtained for the intraband carrier distributions $n_{j,k}$ ($j = e$ for electrons and h for holes while k denotes a band state with a definite momentum) and the polarization variable p_k:

$$\frac{\mathrm{d}p_k}{\mathrm{d}t} = -\frac{1+i\delta_k}{T_{2,k}}p_k - \frac{i}{2}\Omega_k(n_{\mathrm{e},k} + n_{\mathrm{h},k} - 1), \tag{68}$$

$$\frac{\mathrm{d}n_{j,k}}{\mathrm{d}t} = \Lambda_{j,k} - \frac{n_{j,k}}{T_1} - \frac{n_{j,k} - f_{j,k}}{T_{j,k}} + \frac{i}{2}(\Omega_k^* p_k - \Omega_k p_k^*), \tag{69}$$

where $T_{2,k}$ is the dipole-dephasing time (< 100 fs), $\delta_k = (\omega_k - \omega_0)T_{2,k}$ is the k-dependent detuning, and $T_{j,k}$ is the intraband carrier-relaxation time ($\backsim 100$ fs). Individual band states are pumped through the pump term $\Lambda_{j,k}$. The transition frequency ω_k is the renormalized frequency for each individual transition obtained from:

$$\hbar\omega_k = U_{\mathrm{e},k} + U_{\mathrm{h},k} + U_{\mathrm{g}} + \Delta U_k, \tag{70}$$

where $U_{\mathrm{e},k}$ and $U_{\mathrm{h},k}$ are the electron and hole energies, U_{g} is the band-gap energy, and ΔU_k accounts for the band-gap renormalization that occurs because of the many-body effects and depends on the carrier density [60]. The optical field is also modified by the Coulomb effects and is represented by the generalized Rabi frequency:

$$\Omega_k \equiv \frac{\mu_k A}{\hbar} + \frac{1}{\hbar}\sum_{k' \neq k} V_{|k-k'|}p_{k'}, \tag{71}$$

where A is the slowly varying amplitude of the optical field, $V_{|k-k'|}$ is the screened Coulomb potential in the Fourier space, and μ_k is the transition-dipole

moment obtained from band-structure calculations. By solving the semiconductor Bloch equations, Eqs. (68) and (69), in the Fourier domain with the Padé approximation, the optical susceptibility is found to be [59]:

$$\chi(N,\omega) = \frac{1}{\hbar\varepsilon_0 n_0^2 V}\sum_k \frac{|\mu_k|^2 (f_{e,k} + f_{h,k} - 1)}{T_{2,k}^{-1} + i(\omega_k - \omega_0 + \omega)} Q_k, \tag{72}$$

where N is the total carrier density, V is the volume of the semiconductor gain medium, and n_0 is the background refractive index. It is assumed that electrons and holes obey the Fermi–Dirac statistics with distribution functions $f_{e,k}$ and $f_{h,k}$, respectively. The many-body effects enhance the susceptibility through the Coulomb-enhancement factor Q_k [60].

3.1.1. Macroscopic Maxwell–Bloch equations

Numerical studies based on the semiconductor Bloch equations require substantial computing resources, even when the problem is simplified by making several approximations [61, 62]. This is why considerable effort has been directed in recent years to derive simple macroscopic models from the semiconductor Bloch equations that capture the essential features of the microscopic dynamics [63, 68]. Two strategies have been used to arrive at equations for the macroscopic carrier density N and the macroscopic polarization $\mathcal{P}$. In one approach [63–65], the microscopic quantities $n_{j,k}$ and p_k are summed over all k-states. As is generally the case for inhomogeneously broadened systems [69], this summation leads to an infinite hierarchy of coupled equations for the polarization. This hierarchy can be truncated in several different ways [63, 64], yielding different macroscopic models. In the second approach, the susceptibility χ in Eq. (72) is approximated in various ways [66, 68]. Fitting of χ with multiple Lorentzian profiles yields good results for optical pulses wider than a few picoseconds [68]. Recently, such a model was used to simulate the performance of an MOPA (master oscillator/power amplifier) device [70].

We focus here on a specific macroscopic model [64] that leads to a set of generalized Lorenz–Haken equations [67], and thus, allows us to make a direct link to the results of Section 2. We neglect the many-body effects leading to the Coulomb-terms $V_{|k-k'|}$ and Q_k in Eqs. (71) and (72). The macroscopic quantities N and B are defined as [see Eq. (4) for the definition of B]:

$$N = \frac{1}{V}\sum_k n_{e,k} \equiv \frac{1}{V}\sum_k n_{h,k}, \tag{73}$$

$$B = \frac{1}{V}\sum_k (\mu_k p_k + \text{c.c.}). \tag{74}$$

By integrating Eq. (69) with the appropriate density of states, we obtain the carrier-density equation:

$$\frac{\mathrm{d}N}{\mathrm{d}t} = \frac{J}{qd} - \frac{N}{T_1} + \frac{1}{2\hbar}\mathrm{Im}(A^*B), \tag{75}$$

which has the same structure as Eq. (21). The pumping terms $\Lambda_{j,k}$ in Eq. (69) result from the current density J injected into an active layer of thickness d. Note that the intraband relaxation times $T_{j,k}$ do not appear in Eq. (75) since the total carrier density remains unaffected by various intraband scattering processes.

Integration of Eq. (68) over the band states is not straightforward because of the term $\delta_k p_k$, which makes it impossible to get a closed form for $\mathrm{d}P/\mathrm{d}t$. This is a well-known problem for inhomogeneously broadened two-level systems [69], and one ends up with a hierarchy of coupled equations for the variables $\langle \delta_k^m p_k \rangle$, where $m = 1, 2, \ldots$ and $\langle \cdots \rangle$ denotes integration over the band energies. In one approach [63], such a hierarchy is truncated after $m = 2$, resulting in *three* macroscopic equations from the single microscopic Eq. (68). Effects of coherent Coulomb exchange can also be included [65]. A steady-state analysis finds results in agreement with the experimental observations [71, 72]. An alternative way to terminate the hierarchy was proposed in Ref. [64]. If Eq. (68) is divided by $1 + i\delta_k$ before the integration over the band states is performed, and two complex parameters κ and ζ are introduced by using the definitions

$$\kappa N = \left\langle \frac{n_{\mathrm{e},k} + n_{\mathrm{h},k} - 1}{1 + i\delta_k} \right\rangle \tag{76}$$

$$\frac{B}{2\mu\zeta} = \left\langle \frac{p_k}{1 + i\delta_k} \right\rangle, \tag{77}$$

the evolution equation for the macroscopic polarization B becomes:

$$\frac{\mathrm{d}B}{\mathrm{d}t} = -\zeta\left(\frac{B}{T_2} + \frac{i\mu^2}{\hbar}\kappa A N \right). \tag{78}$$

In general, the two complex parameters κ and ζ depend on the carrier density N and the optical intensity $|E|^2$. If the steady state is used to calculate numerically the two parameters above the first laser threshold, κ can be treated as a constant, while ζ depends linearly on N to a good approximation [64].

The field equation is still given by Eq. (29). It is reproduced here for convenience:

$$\frac{\partial A}{\partial z} + \frac{1}{v_0}\frac{\partial A}{\partial t} = \frac{i\beta_0}{2\varepsilon_0 n_0^2} B - \frac{A}{2T_{\mathrm{ph}} v_0}, \tag{79}$$

where T_{ph} accounts for all optical losses. The set of three equations, Eqs. (75), (78) and (79), becomes identical to the Maxwell–Bloch equations obtained for a two-level system in Section 2, if we introduce an effective dipole-dephasing time T_2^{eff} and an effective detuning parameter δ^{eff} as:

$$T_2^{\text{eff}} = T_2 \,\text{Re}(\zeta), \tag{80}$$

$$\delta^{\text{eff}} = \text{Im}(\zeta)/T_2. \tag{81}$$

Although, in principle, these effective two-level parameters need to be determined self-consistently using Eqs. (76) and (77), to a good approximation ζ varies linearly with N in the range over which N is likely to vary in most semiconductor lasers ($N \backsim 1\text{–}5 \times 10^{18}$ cm^{-3}). A linear variation of ζ (although called differently) was also used in Ref. [66].

3.1.2. Generalized Lorenz–Haken equations

As in Section 2.2, we make the mean-field approximation and set $\partial A/\partial z = 0$ in Eq. (79). This approximation is reasonable for semiconductor lasers operating in a single longitudinal mode. When the assumption is made that the parameters κ and ζ can be treated as constants (a reasonable approximation if the semiconductor laser is operating above threshold) and the dynamic variables x, y and z are introduced as in Section 2.2, the macroscopic Maxwell-Bloch equations obtained above can be written in the form of *generalized* Lorenz–Haken equations [67]:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -\sigma(x - y), \tag{82}$$

$$\frac{\mathrm{d}y}{\mathrm{d}t} = -(1 + i\theta)[y - (1 - i\alpha)(r - z)x], \tag{83}$$

$$\frac{\mathrm{d}z}{\mathrm{d}t} = -bz + \text{Re}(x^* y). \tag{84}$$

The two new parameters α and θ are defined as:

$$\alpha = -\text{Im}(\kappa)/\text{Re}(\kappa), \tag{85}$$

$$\theta = \text{Im}(\zeta)/\text{Re}(\zeta). \tag{86}$$

Physically, α governs the coupling between amplitude and phase variations. It is also known as the linewidth-enhancement factor. The effective detuning θ has its origin in the band structure responsible for inhomogeneous broadening. Owing to the self-consistency requirement, both κ and ζ (and, therefore, α and θ) are, in principle, time-dependent through the carrier density. When the semiconductor laser is operating in the CW mode, one can generally ignore this time dependence. Strictly speaking, the dynamics of α and θ should be taken into account when a linear-stability analysis is performed. However, it can be neglected in practice since the carrier lifetime, $T_1 \backsim 1$ ns, is much larger than the photon lifetime, $T_{\text{ph}} \backsim$ 1 ps [64]. Of course, when the system is pumped above the second threshold, the dynamics of α and θ cannot be neglected without further justification.

The Lorenz–Haken equations, Eqs. (35)–(37), of Section 2 are a special case of Eqs. (82)–(84). When $\alpha = \theta = 0$, one readily obtains the standard Lorenz–Haken

equations obtained in the resonant case with $\delta = 0$ [5]. When $\alpha = \theta \neq 0$, Eqs. (82)–(84) reduce to those describing a detuned two-level system.

Eqs. (82)–(84) can be used to study whether a second threshold exists for semiconductor lasers. A linear stability analysis of the trivial solution, $x = y = z = 0$, reveals that the first threshold can depend substantially, but not dramatically, on the values of α and θ. More interesting is the investigation of the second threshold. Typical parameters for a single-mode semiconductor laser lead to $\sigma \backsim 10^{-2}$ and $b \backsim 10^{-4}$. It is immediately obvious that such values of σ and b do not satisfy the bad-cavity condition in Eq. (64). As a result, the resonant Lorenz–Haken model does not predict a second threshold. When analyzing the linear stability of the CW solutions of Eqs. (82)–(84), it was found that a second threshold *does* exist for semiconductor lasers, in a substantial region of the (α, θ) parameter space [67]. Unfortunately, the pump strength required to reach the second threshold is extremely high. If the carrier lifetime, T_1, can be reduced to a

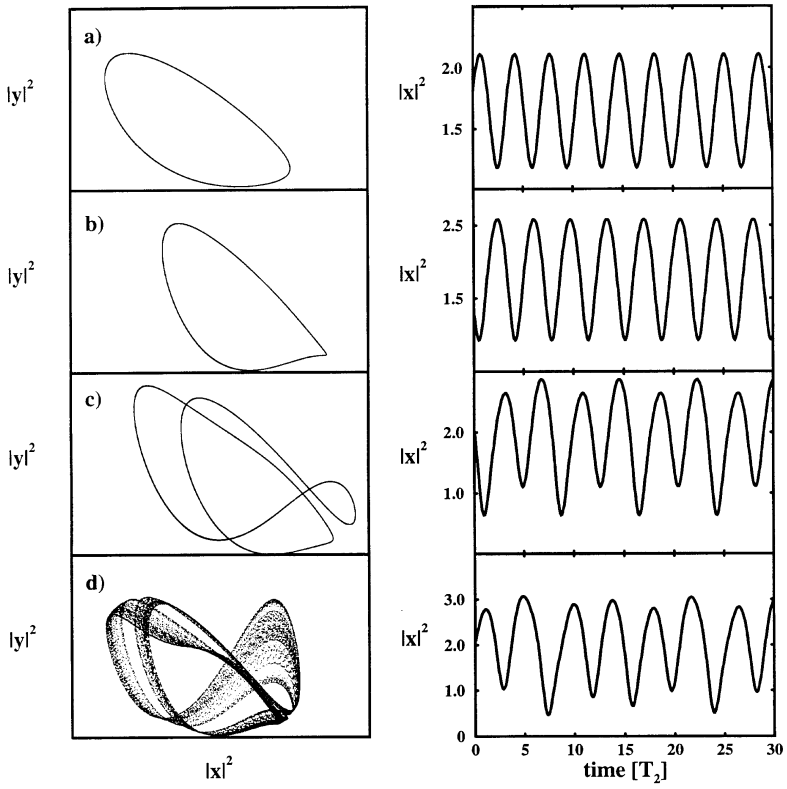


Fig. 5. Phase-space (left column) and temporal (right column) dynamics of a quantum-well semiconductor laser designed with $T_{ph} = 75$ fs and $T_1 = 500$ fs. Parameter values used are: $\sigma = 0.3$, $b = 0.2$, $\alpha = 2$, and $\theta = -1.6$. The first and second thresholds occur at $r_{th}^{(1)} = 1.006$ and $r_{th}^{(2)} = 8.9$, respectively. The four sets of curves correspond to different values of the ratio $r/r_{th}^{(2)} = 1.05$ (a), 1.15 (b), 1.2 (c) and 1.235 (d).

picosecond range, the second threshold reduces enough to become practical. Fig. 5 shows the dynamics of such an engineered quantum-well laser at different pump levels. Note that this laser still does not satisfy the bad-cavity condition.

3.2. Semiconductor-laser rate equations

For most semiconductor lasers, the polarization dynamics can be safely eliminated adiabatically unless they are operated in the femtosecond regime. We then end up with two first-order ordinary differential equations that have become known as the semiconductor-laser rate equations. They have proven to be quite adequate in describing laser dynamics under most practical conditions and have the advantage that considerable analytical progress can be made in understanding the laser dynamics.

3.2.1. Rate-equation approximation

In the rate-equation approximation, adiabatic elimination of the material polarization is carried out by setting $dB/dt = 0$ in Eq. (78), resulting in:

$$B = -i(\mu^2 T_2/\hbar)\kappa A N. \tag{87}$$

By substituting this expression in Eqs. (75) and (79) we can eliminate B and obtain the following set of two equations:

$$\frac{dA}{dt} = \frac{1}{2}G(N)A - \frac{A}{2T_{ph}}, \tag{88}$$

$$\frac{dN}{dt} = \frac{J}{qd} - \frac{N}{T_1} - \frac{\varepsilon_0 n_0^2}{2\hbar\omega_0}\mathrm{Re}[G(N)] \mid A \mid^2, \tag{89}$$

where

$$G(N) = \frac{\mu^2\omega_0 T_2}{\varepsilon_0\hbar n_0^2}\kappa(N)N. \tag{90}$$

The quantity $G(N)$ is related to the material gain, except that it is complex because the parameter κ is generally complex. As already mentioned, κ depends only weakly on the carrier density N above the first laser threshold. For 'bulk' semiconductor lasers (active-layer thickness $\gg$10 nm), it is usually sufficient to use a linear functional form for $G(N)$, while for quantum-well lasers a logarithmic dependence of $G(N)$ on N is more appropriate [73]. For all semiconductor lasers, we can linearize $G(N)$ around the first laser threshold and use

$$G(N) = G(N_{th}) + (1 - i\alpha)G_N(N - N_{th}), \tag{91}$$

where G_N is the differential-gain parameter and N_{th} is the threshold carrier density. The quantity $G(N_{th})$ is generally complex. However, by a suitable choice of the carrier frequency ω_0, the imaginary part of $G(N_{th})$ can be reduced to zero;

$G(N_{\mathrm{th}})$ can then be interpreted as the threshold value of the gain. The parameter α in Eq. (91) is the linewidth-enhancement factor introduced earlier [74]. For most semiconductor lasers, its value is in the range 2–8. One can not overstress the importance of this parameter and its influence on the laser dynamics. It can be shown through a noise analysis that the linewidth of the laser mode is enhanced by a factor of $1 + \alpha^2$ [55]. This enhancement was already predicted in 1967 by both Lax [75] and Haug and Haken [76], but was thought to be negligible in practice.

Substituting Eq. (91) in Eqs. (88) and (89) and noting that $G(N_{\mathrm{th}}) = T_{\mathrm{ph}}^{-1}$ beyond the first laser threshold, we obtain the standard rate equations for single-mode semiconductor lasers:

$$\frac{\mathrm{d}A}{\mathrm{d}t} = \frac{1}{2}(1 - i\alpha)G_{\mathrm{N}}(N - N_{\mathrm{th}})A, \tag{92}$$

$$\frac{\mathrm{d}N}{\mathrm{d}t} = \frac{J}{qd} - \frac{N}{T_1} - [T_{\mathrm{ph}}^{-1} + G_{\mathrm{N}}(N - N_{\mathrm{th}})]P, \tag{93}$$

where $P = (\varepsilon_0 \mathrm{n}_0^2/2\hbar\omega_0)|A|^2$ denotes the photon density within the laser cavity.

Several approximations were made in driving these two rate equations. For example, the effects of carrier diffusion have been neglected. This neglect is justified for single-mode lasers in which the active-layer width is not too large compared with the diffusion length. Diffusion must be included for broad-area semiconductor lasers. Another approximation is that the carrier lifetime T_1 is assumed to be constant. In practice, T_1 can depend on N if electron–hole recombination mechanisms such as the Auger effect are taken into account. Such mechanisms can be included by using the following functional form for T_1 [77]:

$$T_1^{-1}(N) = \tau_{\mathrm{nr}}^{-1} + BN + CN^2, \tag{94}$$

where τ_{nr}^{-1} accounts for nonradiative recombinations at defects or impurities within the active layer, BN accounts for spontaneous radiative recombinations, and CN^2 is due to the Auger-recombination processes. For an overview of different types of Auger processes, see Ref. [55]. Since $N \approx N_{\mathrm{th}}$ above the first laser threshold, one can consider the carrier lifetime to be a constant by using $T_1 \equiv T_1(N_{\mathrm{th}})$.

Since A is generally complex, it is useful to introduce the optical phase φ through

$$A(t) = \sqrt{P(t)}\exp[i\varphi(t)]. \tag{95}$$

we then obtain the following three rate equations for single-mode semiconductor lasers:

$$\frac{\mathrm{d}P}{\mathrm{d}t} = G_{\mathrm{N}}(N - N_{\mathrm{th}})P, \tag{96}$$

$$\frac{dN}{dt} = \frac{J}{qd} - \frac{N}{T_1} - [T_{ph}^{-1} + G_N(N - N_{th})]P, \tag{97}$$

$$\frac{d\varphi}{dt} = -\frac{1}{2}\alpha G_N(N - N_{th}). \tag{98}$$

However, since $\varphi(t)$ does not appear in the first two equations and just follows the carrier density, the phase is a slaved variable in the language of the nonlinear dynamics. The dynamics of a single-mode semiconductor laser is, thus, governed by the two first-order differential Eqs. (96) and (97) for the photon density P and the carrier density N. We turn to their solutions in the next section.

3.2.2. Relaxation oscillations

We first consider the stability of the steady-state solutions associated with Eqs. (96) and (97). As before, the trivial solution, $P = N = 0$, becomes unstable when the first threshold is reached at $J = J_{th} \equiv qdN_{th}/T_1$. When J is larger than this threshold value, the steady-state CW solution of Eqs. (96) and (97) is given by:

$$P_s = (J - J_{th})T_{ph}/qd, \tag{99}$$

$$N_s = N_{th}. \tag{100}$$

A linear stability analysis of this CW solution shows that small perturbations from the steady state evolve as $\exp(st)$ with the growth rate

$$s_{1,2} = -\lambda_R \pm i\sqrt{\omega_R^2 - \lambda_R^2} \tag{101}$$

where

$$\lambda_R = \frac{1}{2}\left(\frac{1}{T_1} + G_N P_s\right) \tag{102}$$

$$\omega_R = \sqrt{G_N P_s / T_{ph}}. \tag{103}$$

The two complex conjugate values for s describe an oscillatory decay of perturbations. Such oscillations are known as *relaxation oscillations* and are characterized by the damping rate λ_R and the angular frequency $\Omega_R = \sqrt{\omega_R^2 - \lambda_R^2}$. Since perturbations from the steady state always decay exponentially for all values of $J > J_{th}$, the CW state is stable, and the second threshold does not exist for semiconductor lasers whose dynamics is governed by the above two rate equations. This is not surprising since the rate equations, Eqs. (96) and (97), describe a class-B laser.

Several factors can affect the above conclusion. A simple way to make a semiconductor laser unstable is to introduce a saturable absorber within the laser cavity. Owing to the monolithic nature of the semiconductor-laser cavity, it is not obvious how to do so. One scheme injects the current only over a portion of the

laser cavity and applies a reverse bias over the remaining part. The gain is positive only over the section where electron–hole pairs are injected, and the reverse-biased section acts as a saturable absorber. One can model the effect of saturable absorption approximately by making the gain G in Eq. (91) a function of both N and P. If we expand $G(N, P)$ in a Taylor series around the threshold value $P = 0$, $G(N, P) = G(N, 0) + G_P P$, where $G(N, 0)$ is given by Eq. (91), and G_P is a constant. This change amounts to adding a term of the form $G_P P^2$ to the right side of Eq. (96) [55]. It is a simple exercise to show that the decay rate of relaxation oscillations is then given by:

$$\lambda_R = \frac{1}{2}\left(\frac{1}{T_1} + G_N P_s - G_P P_s\right), \tag{104}$$

and can become negative for sufficiently large values of G_P, leading to a second laser threshold above which self-pulsing can occur.

There are also mechanisms that increase the decay rate λ_R of relaxation oscillations in a semiconductor laser, making the laser more stable. Through many experimental studies it has become clear that inclusion of the power dependence of the gain $G(N, P)$ is of crucial importance for describing the dynamics accurately far above the first laser threshold [78]. At high power levels, processes like spectral hole burning and carrier heating effectively cause the gain to decrease with an increase in P [55]. Although the exact functional form of $G(N, P)$ depends on the physical mechanism involved, to a good approximation the nonlinear gain effect can be described by using a simple form:

$$G(N, P) = G(N, 0)(1 - \varepsilon P), \tag{105}$$

since εP is usually less than a few percent. A straightforward linear stability analysis [78] shows that this small contribution has a large impact on the decay rate of relaxation oscillations, which becomes:

$$\lambda_R = \frac{1}{2}\left(\frac{1}{T_1} + G_N P_s + \frac{\varepsilon P_s}{T_{ph}}\right). \tag{106}$$

Since $T_{ph}/T_1 \backsim 10^{-3}$ in most semiconductor lasers, λ_R increases by a factor of 2 even if εP_s is below 1%.

Since the optical phase is a slaved dynamical variable, the semiconductor-laser dynamics is governed by only two dynamical variables, namely the power $P(t)$ and the carrier density $N(t)$. It is well known that at least three dynamical variables are needed for a nonlinear system to exhibit instabilities. The injection of external optical signals into the laser cavity causes the optical phase to become the third dynamical variable, suggesting that new instabilities may arise. The effects of external signals, through optical injection or feedback, on the dynamics of semiconductor lasers, has been a hot topic for a number of years. We turn to this topic next.

3.3. Optical injection

In this section, we consider the practically important case of optical injection in which the semiconductor laser is injected with light from an external laser source (often called the master laser). Since the dynamics of the semiconductor laser is then influenced by the master laser, it is referred to as the 'slave' laser. The rate Eq. (92) for the optical field has an injection-induced additional term and takes the form:

$$\frac{\mathrm{d}A}{\mathrm{d}t} = \frac{1}{2}(1 - i\alpha)G_{\mathrm{N}}(N - N_{\mathrm{th}})A + \tau_{\mathrm{L}}^{-1}\sqrt{P_x}\exp(-ivt), \tag{107}$$

where τ_{L} is the round-trip time in the laser cavity, P_x is the power of the external signal injected into the single mode of the laser, and $v = \omega_x - \omega_0$ accounts for the frequency difference between the slave laser (frequency ω_0) and the master laser (frequency ω_x).

Eqs. (97) and (107) describe the dynamics of a slave semiconductor laser in response to the injected signal from a master laser. Depending on the strength of the injected signal, the slave laser can either change its frequency of operation to that of the master laser, and, thus, *lock* its frequency to ω_x, or engage in a more complicated dynamics in response to the external signal. As discussed below, both of these scenarios are predicted by the modified rate equations.

When the slave laser locks its frequency to the injected field, the steady-state CW solution of Eqs. (97) and (107) can be written as:

$$A(t) = \sqrt{P_{\mathrm{L}}}\exp[-i(vt + \varphi_{\mathrm{L}})], \quad N(t) = N_{\mathrm{th}} + n_{\mathrm{L}}, \tag{108}$$

where φ_{L} accounts for the (locked) phase difference between the master and slave lasers. A necessary condition for frequency locking to occur is obtained by substituting Eq. (108) in Eqs. (97) and (107), and is found to be:

$$|\, v \,| \leq v_{\mathrm{L}} \equiv \frac{\sqrt{1 + \alpha^2}}{\tau_{\mathrm{L}}}\sqrt{\frac{P_x}{P_{\mathrm{L}}}}. \tag{109}$$

When this condition is fulfilled and the slave laser is operating with $J > J_{\mathrm{th}}$, there are two CW states of the form of Eq. (108), both of which have the same intensity P_{L}, but different phases φ_{L}. The linear stability analysis shows that only one of these CW states can be stable (depending on the parameter values), while the other one corresponds to a saddle point in the phase space [31, 79]. The stable solution can be destabilized either by increasing the ratio P_x/P_0, where P_0 is the output of the slave laser in absence of injection, or by increasing the detuning v between the slave and master frequencies. Fig. 6 shows the different regimes of operation in the two-dimensional parameter space formed by using v and P_x. There are three regimes of operation, which we label as stable locking (LS), destabilized locking (LNS) and non-locking (NL). Notice that for a negative detuning v, stable frequency locking is always possible, in contrast with the case of

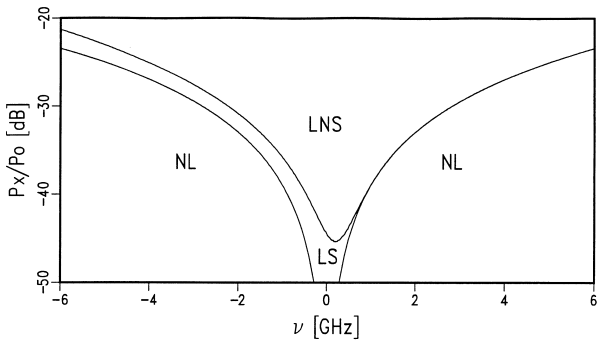


Fig. 6. Different operating regimes in the (v, P_x) parameter space for an optically injected semiconductor laser. The parameters v and P_x denote the frequency detuning and the power of the injected signal. LS marks the stable locking region, LNS marks the unstable locking region, and NL marks the regime in which injection locking does not occur.

positive detunings. This is a consequence of the carrier-induced changes governed by the α parameter.

In the NL regime, the slave laser manages not to lock on the frequency dictated by the master laser. However, because of the presence of a third frequency in the nonlinear system, the relaxation-oscillation frequency ω_R, interesting wave-mixing effects still take place in the NL regime [80–82]. The closer to the boundary between NL and LS (or LNS) regions the system operates, the stronger the wave-mixing effects become [79]. Far away from that boundary, one can describe the dynamics satisfactorily through a four-wave-mixing process [80, 81].

The parameter space depicted in Fig. 6 contains a wealth of interesting nonlinear dynamics. In 1992 Sacher et al. [83] carried out one of the first experimental investigations of the injection-induced effects from a nonlinear-dynamics point of view. Since then, several theoretical papers have discussed the instabilities and chaotic features associated with optical injection [84–90]. Lee et al. [84] found a period-doubling route to chaos. Annovazzi-Lodi et al. [85] showed that an intermediate chaotic region exists in between the NL and LS regimes. Simpson et al. [86] found period-doubling cascades with changes in the injection level. Kovanis et al. [87] mapped out various instabilities experimentally, and found two islands of chaos separated by regions of period-1 and period-2 solutions. Erneux et al. [88] derived, in 1996, a third-order pendulum equation that describes several aspects of the bifurcation line (the boundary between the LNS–NL regime). De Jagher et al. [89] investigated bifurcations of the relaxation oscillation in the locking region by using a two-variable scalar function, similar to a 'thermodynamic' potential. Analytical expressions for the stability boundaries have also been obtained using asymptotic techniques [90].

3.4. Optical feedback

As a result of their low facet reflectivities, semiconductor lasers are extremely sensitive to spurious reflections, which cause a fraction of the light emitted by the laser to re-enter the laser cavity. Surprisingly, feedback levels of less than −40 dB (0.01%) routinely cause dramatic changes in the laser output. Although the effects of optical feedback on the operation of a semiconductor laser were studied earlier [91], the 1980 paper by Lang and Kobayashi [92] is generally considered a milestone in the sense that it initiated an enormous research effort devoted to studying the optical-feedback effects. Some of these studies consider not only CW but also the self-pulsing operation of a semiconductor laser and show that the self-pulsing frequency can be locked to the round-trip frequency associated with the external cavity, making the repetition rate of the pulse train tunable [35, 93].

Lang and Kobayashi showed that a semiconductor laser, when subjected to external optical feedback, can show multistability as well as hysteresis features [92], analogous to those occurring in a nonlinear Fabry–Perot resonator. The presence of a reflecting surface outside the laser cavity creates an external cavity, which has its own longitudinal modes with a frequency separation $\Delta\nu_{\mathrm{ext}} = 1/\tau_{\mathrm{fb}}$, τ_{fb} being the round-trip time in the external feedback cavity. The existence of two sets of longitudinal modes leads to a competition between the laser cavity (with a much larger mode spacing) and a passive cavity (no gain in the external cavity), each having its own resonances. As a result of the α parameter, the semiconductor laser is able to change its frequency to accommodate external-cavity resonances. As a result of this feature, a whole set of external-cavity mode frequencies becomes available for lasing action, each with different stability properties [31].

External feedback also affects the laser noise considerably. Depending on the feedback conditions, the linewidth of the laser mode resulting from phase fluctuations may increase or decrease [94]. In fact, by a proper phase matching of the feedback signal, the linewidth can be reduced by more than a factor of 10 [95–97]. By changing the feedback parameters slightly, multistability has been observed in the laser output, as the laser performs 'mode-hops' from one external cavity mode to another [94]. A 'thermodynamic-potential' model for phase diffusion has been proposed to describe this mode-hopping phenomenon. In this model, the instantaneous laser frequency undergoes Brownian motion in a potential landscape containing multiple minima, each minimum pertaining to an external cavity mode [98, 99].

Although the main focus during the early 1980 s was on noise-related issues [100], attention gradually shifted toward the deterministic dynamics induced by optical feedback. Bistability, self-pulsing and chaotic emission were observed as early as 1983 from a GaAs laser coupled to an external cavity [101, 102]. The chaotic state characterized by a dramatic linewidth broadening (typically from 100 MHz to 25 GHz) was baptized in 1985 as *coherence collapse* [103]. Since then, this highly complicated dynamical state has been studied extensively and has proven to contain a wealth of interesting nonlinear dynamics. Under similar

feedback conditions, but usually at lower pump currents, the phenomenon of low-frequency fluctuations (LFF) is found to occur. The LFFs are characterized by apparently random power drop-outs and have been identified as the origin of an undesirable 'kink' in the light-current (L–I) curve [104]. Although models explaining LFF have been formulated since 1986 [105], it is only recently that an experiment has shown conclusively that the mechanism underlying LFF is deterministic; it has been termed *chaotic itinerancy with a drift* [106].

3.4.1. Lang–Kobayashi equations

Feedback-induced nonlinear dynamics can be studied by using the Lang–Kobayashi equations, which are nothing but the rate Eqs. (92) and (93) modified to account for the optical feedback:

$$\frac{\mathrm{d}A}{\mathrm{d}t} = \frac{1}{2}(1 - i\alpha)\, G_{\mathrm{N}}(N - N_{\mathrm{th}})A + \gamma_{\mathrm{fb}} A(t - \tau_{\mathrm{fb}})\exp(i\omega_0 \tau_{\mathrm{fb}}), \tag{110}$$

$$\frac{\mathrm{d}N}{\mathrm{d}t} = \frac{J}{qd} - \frac{N}{T_1} - [T_{\mathrm{ph}}^{-1} + G_{\mathrm{N}}(N - N_{\mathrm{th}})]P. \tag{111}$$

Eq. (110) assumes feedback to be so weak, that multiple round-trip effects in the external cavity can be neglected. The parameter γ_{fb} is called the feedback rate and is related to the facet reflectivity R_1 and the external reflectivity R_{ext} by [31]:

$$\gamma_{\mathrm{fb}} = \eta_{\mathrm{c}} \frac{1 - R_1}{\tau_{\mathrm{L}}} \sqrt{\frac{R_{\mathrm{ext}}}{R_1}}, \tag{112}$$

where the coupling efficiency η_{c} accounts for the effects that make the coupling back into the laser less than 100% effective. As before, we can treat T_1 as a constant if the feedback-induced changes in the carrier density are relatively small. Stochastic noise should be added if the rate equations are used to study phenomena such as spontaneous-emission-induced mode hopping [98, 99] and feedback-induced linewidth reduction. In this review, we focus on the deterministic effects and, thus, do not include spontaneous-emission noise.

3.4.2. Steady-state solutions

The steady-state CW solutions of Eqs. (110) and (111) can be written as:

$$A(t) = \sqrt{P_{\mathrm{s}}}\exp(-i\Delta\omega_{\mathrm{s}} t), \quad N(t) = N_{\mathrm{th}} + n_{\mathrm{s}}, \tag{113}$$

where $\Delta\omega_{\mathrm{s}}$ is the frequency shift and n_{s} is the carrier-density change induced by the feedback. Multiple solutions are found to exist and correspond to different values of $\Delta\omega_{\mathrm{s}}$ and n_{s}. These values lie on an ellipse in the ($\Delta\omega_{\mathrm{s}}$, n_{s}) plane [105]:

$$\left(\frac{G_{\mathrm{N}} n_{\mathrm{s}}}{2\gamma_{\mathrm{fb}}}\right)^2 + \left(\frac{\Delta\omega_{\mathrm{s}}}{\gamma_{\mathrm{fb}}} - \frac{\alpha G_{\mathrm{N}} n_{\mathrm{s}}}{2\gamma_{\mathrm{fb}}}\right)^2 = 1. \tag{114}$$

The steady-state values of $\Delta\omega_{\mathrm{s}}$, n_{s}, and P_{s} are given by:

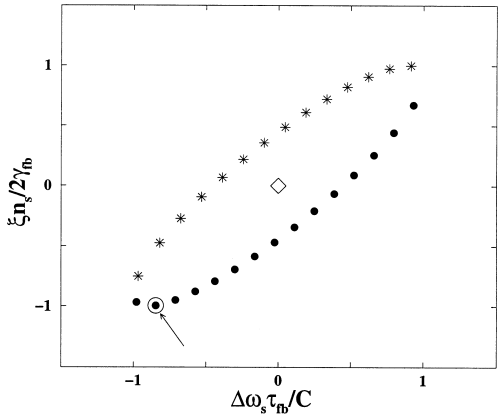


Fig. 7. Location of the steady-state solutions (fixed points) in the ($\Delta\omega_s$, n_s) parameter space, where $\Delta\omega_s$ is the frequency shift and n_s is the carrier-density change for each solution. Parameter values used are $\alpha = 2$ and $\omega_0\tau_{fb} = 0$. The diamond at the center denotes the only steady-state solution remaining in the absence of feedback. Stars correspond to antimodes (saddle points) while circles indicate 'normal' compound-cavity modes. The arrow points at the minimum-threshold state that benefits maximally from the feedback.

$$\Delta\omega_s = -\frac{C}{\tau_{fb}}\sin[(\omega_0 + \Delta\omega_s)\tau_{fb} + \tan^{-1}\alpha], \tag{115}$$

$$n_s = -\frac{2\gamma_{fb}}{G_N}\cos[(\omega_0 + \Delta\omega_s)\tau_{fb}], \tag{116}$$

$$P_s = \frac{(J - J_{th})/qd - n_s/T_1}{T_{ph}^{-1} + G_N n_s}, \tag{117}$$

where C is a dimensionless feedback parameter defined as [31]:

$$C = \gamma_{fb}\tau_{fb}\sqrt{1 + \alpha^2}. \tag{118}$$

Fig. 7 shows a typical set of steady-state solutions (circles and stars) with the diamond at the center indicating the only CW solution of the laser in the absence of feedback. Each steady-state solution is characterized by a specific combination of the mode frequency $\omega_0 + \Delta\omega_s$, the associated carrier density $N_{th} + n_s$, and the resulting power P_s. By noting from Eq. (117) that the smallest value of n_s will result in the highest output power P_s, the fixed point with the least carrier density is called the 'minimum-threshold' state and is indicated with an arrow in Fig. 7. Operating in that state, the laser benefits maximally from the feedback, i.e. there is optimal constructive interference between the intracavity field and the feedback field. In view of the important role of the relative phase difference between the two fields, the proper dynamical variables are the round-trip phase difference $\eta(t) = \varphi(t) - \varphi(t - \tau_{fb})$, the photon density $P(t)$, and the excess carrier density $n(t)$.

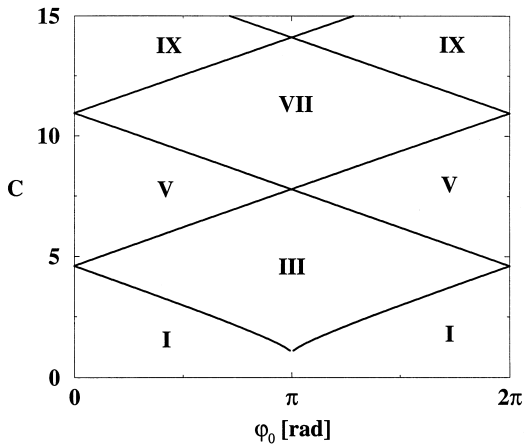


Fig. 8. Boundaries in the (C, φ_0) parameter space showing that a semiconductor laser can operate in multiple compound-cavity modes as the amount of feedback increases. The number of possible modes (fixed points) are indicated by Roman numerals.

The number of *possible* lasing states is determined by Eq. (115) which contains only two dimensionless parameters, the feedback rate C and the feedback phase $\varphi_0 \equiv \omega_0 \tau_{\mathrm{fb}} + \tan^{-1}\alpha$. Each compound-cavity mode frequency $\omega_0 + \Delta\omega_{\mathrm{s}}$ causes a different change n_{s} in the carrier density as expressed by Eq. (116). When the pump rate J is sufficiently large [see Eq. (117)], several compound-cavity modes can be excited simultaneously. Fig. 8 shows the number of possible lasing states in the (φ_0, C) plane. The curves in Fig. 8 are obtained by using the relation:

$$\varphi_0 = (2k+1)\pi \pm \cos^{-1}\left(\frac{1}{C}\right) \mp C \sin\left[\cos^{-1}\left(\frac{1}{C}\right)\right], \tag{119}$$

where $C \geqslant 1$ and k is an integer. Clearly, the line $C = 1$ is a special one: when $C < 1$ the laser has only one frequency to choose from, which is only slightly shifted from the solitary-laser frequency ω_0. As the feedback level increases, additional compound-cavity modes are created in pairs. It turns out that the lines given by Eq. (119) indicate the location of a *saddle-node bifurcation* [31].

3.4.3. Linear-stability analysis

The linear-stability analysis of the steady-state solutions of the Lang–Kobayashi equations results in a rather complicated characteristic equation that is not a polynomial equation, because of the feedback delay time τ_{fb} [31]. It can be shown relatively easily that *half* of the fixed points in the phase space (denoted by stars in Fig. 7) are saddle points. Physically, these fixed points correspond to the situation in which feedback is out of phase, causing destructive interference. They are characterized by the condition [107, 108]:

$$1 + C\cos[(\omega_0 + \Delta\omega_{\mathrm{s}})\tau_{\mathrm{fb}} + \tan^{-1}\alpha] > 0 \tag{120}$$

and lie in the upper half of the ellipse shown in Fig. 7. These fixed points are sometimes referred to as the 'antimodes' of the laser [109].

The fixed points lying on the lower half of the ellipse in Fig. 7 (shown by circles) may or may not be stable against small perturbations. To investigate their stability, one needs to find how relaxation oscillations evolve with time. Exact analytical expressions for the frequency and the decay rate of relaxation oscillations are hard to obtain [110, 114], but approximate expressions have been derived. The minimum feedback rate required for a Hopf instability to occur is found to be [113]:

$$\gamma_{\mathrm{fb}}^{\min} \simeq \frac{\lambda_{\mathrm{R}}}{\sqrt{1+\alpha^2}}, \tag{121}$$

where λ_{R} is the damping rate of relaxation oscillations in the absence of feedback and is given by Eq. (102).

Recently, asymptotic techniques have been used to obtain expressions for the loci of the Hopf bifurcation [115]. Although these expressions are approximate, the error is quantifiable. Fig. 9 shows the Hopf-bifurcation lines in the (φ_0, C) plane. By increasing the feedback strength C and keeping the feedback phase φ_0 fixed, one encounters a series of Hopf- and saddle-node bifurcations. For instance when $\varphi = 0.9\pi$, the single compound-cavity mode loses its stability as feedback is increased beyond $C \backsim 2$. Before that happens, two compound-cavity modes are created through a saddle-node bifurcation at $C \backsim 1.5$. One of these is a saddle point, while the other loses its stability at $C \backsim 3.2$. Then, until the next saddle-node bifurcation occurring near $C = 7.5$, the system is characterized by two destabilized compound-cavity modes and one antimode.

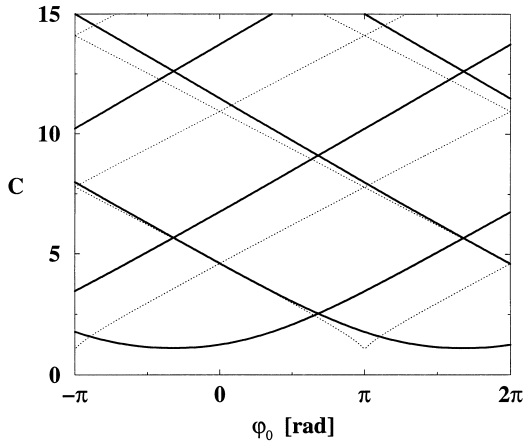


Fig. 9. Boundaries in the (C, φ_0) parameter space showing the location of Hopf instabilities (solid curves). Dotted lines reproduce the boundaries of Fig. 8 for comparison. Parameter values used are: $\alpha = 4$, $\tau_{\mathrm{fb}} = 4$ ns, $T_{\mathrm{ph}} = 0.32$ ps, $G_{\mathrm{N}} = 5625\ \mathrm{s}^{-1}$, $T_1 = 2.2$ ns, $N_{\mathrm{th}} = 2.14 \times 10^8$, and $J/J_{\mathrm{th}} = 1.5$.

Each Hopf bifurcation corresponds to a transition from the CW to a self-pulsing state (fixed-point to limit-cycle transition in the phase space) at the feedback level at which the Hopf bifurcation occurs. It does not reveal any information about the nonlinear dynamics that will occur if the limit cycle becomes unstable at higher feedback levels. The analytical investigation of the instability of limit cycles by means of a Floquet analysis [114], appears at the moment the farthest any analytical technique has gone for the feedback problem.

A thorough understanding of the complicated nonlinear dynamics that is exhibited by a semiconductor laser under optical feedback, therefore, relies on detailed numerical modeling and experimental investigations. However, the results from a linear-stability analysis of the fixed points and limit cycles are indispensable for understanding the experimentally observed behavior. As an example, we mention the models used to study the effect of excited relaxation oscillations on the spontaneous-emission-induced mode hopping [98, 99]. Those models start from the assumption that the fixed points (the compound-cavity modes) are stable and that spontaneous-emission noise will cause random hopping among these modes. Analytical expressions for the power spectrum in that situation have been obtained [117], but the theoretical predictions do not agree too well with the experiments. Numerical simulations showed that, very often, at least one of the fixed points is unstable and is replaced by a limit cycle. A nonlocal 'thermodynamic' potential model has been developed [118] to describe the spontaneous-emission-induced hopping between the limit cycles, and shows good agreement with the recent experiments [119].

3.4.4. Coherence collapse

The dynamical state of coherence collapse is reached when the laser is pumped well above its first threshold ($J/J_{\mathrm{th}} > 2$), and the feedback level is $\backsim 10^{-4}$ (or -40 dB). The term coherence collapse refers to the large increase in the laser-mode linewidth (typically from 100 MHz to 25 GHz) observed when a semiconductor laser is exposed to moderate amounts of feedback from a reflector placed at a distance $\backsim 1$ m. Coherence collapse constitutes a rather complicated dynamical state for a semiconductor laser with hundreds of destabilized compound-cavity modes (and hundreds of antimodes or saddle points). A large number of numerical and experimental results have been published, concerning the chaotic nature of the nonlinear dynamics in the coherence-collapse regime.

The first clear evidence of chaos in semiconductor lasers induced by weak optical feedback dates back to 1990 [120]. In that experiment, the laser underwent a quasi-periodic route to chaos, interrupted sometimes by frequency locking, while spontaneous-emission-induced transitions were found to occur between two coexisting attractors. The latter behavior was already predicted numerically by modeling the system in terms of injection locking [121]. A detailed theoretical and experimental treatment of these phenomena was given later in 1992 [122]. Analytical work based on a Floquet analysis of the limit-cycle behavior [114] substantiated these findings. Another experiment in 1993 observed both the

period-doubling and the quasi-periodic routes to chaos [123]. The period-doubling route is followed by the laser whenever the relaxation oscillation frequency is locked to an external-cavity resonance. In the absence of such frequency locking, the quasi-periodic route to chaos is observed [124]. Whether the laser will follow a period-doubling or quasi-periodic route depends on the length of the external cavity; only a quasi-periodic route is found to occur for long lengths. This was shown to be the case for both multimode [125] and single-mode lasers [126]. A dimensional calculation shows the fractal nature of the coherence-collapse attractor [127]. The inclusion of nonlinear gain through the εP term in Eq. (105) changes the dynamical features considerably, but the fractal nature of the coherence-collapse attractor remains unchanged [128, 129]. The coexistence of several attractors is shown numerically to emanate from different compound-cavity modes bifurcating to different unstable tori, and these tori undergo different types of quasiperiodic routes. When all tori are unstable, intermittency and hyper-chaos are found to occur [130].

3.4.5. Low-frequency fluctuations

When the semiconductor laser is exposed to external feedback but is pumped close to its first threshold, a low-frequency peak (near $\backsim$10 MHz) appears in the power spectrum. Such low-frequency fluctuations (LFFs) are characterized by a gradual build-up of the laser power (over a duration $\backsim 10\ \tau_{fb}$), followed by a sudden drop in the power to the near-zero level, and the process starts all over again. The LFF phenomenon was studied as early as 1977 [91]. An explanation for LFFs was proposed in 1986 [105], and the results of that model agreed qualitatively well with the experimental data [131]. In this model, the statistics of the time interval between successive power drop-out events is described stochastically, with the system making spontaneous-emission-induced escapes over a potential well.

Since crucial assumptions in the 1986 stochastic model [105] of LFFs do not have a clear theoretical basis, research efforts continued to find the physical origin of LFFs. In 1988, a spectral iteration scheme was designed, following up on the notion of noise-induced switching and leading to a dynamical bistability between the minimum threshold state and a temporarily stable low-intensity state [104]. This scheme works well during the initial stages after a drop-out, but fails to predict the long-term behavior. It was also suggested in Ref. [132] that LFFs may correspond to a chaotic attractor in the phase space, rather than being stochastic in nature. This notion was experimentally confirmed in 1989 [132], and the LFF phenomenon was classified as *time-inverted type-II intermittency*. Only in 1994 was the origin of LFFs revealed by Sano [109], who showed that the standard Lang–Kobayashi equations exhibit LFFs as a result of merging of the attractor ruin of a compound-cavity mode with an antimode. This behavior is shown in Fig. 10. Experimentally, such scenario proved very hard to verify until Van Tartwijk et al. [133] showed theoretically that LFFs in fact consist of irregular picosecond pulses, before and after the subsequent crises. Such an irregular pulsing behavior

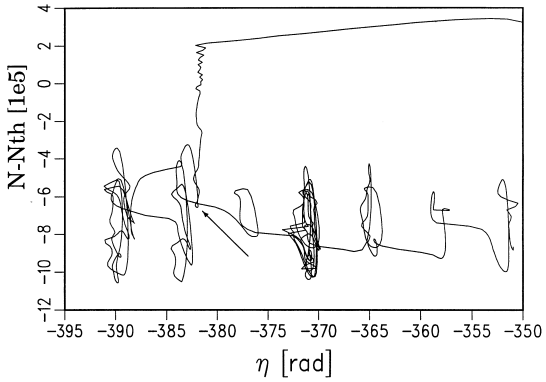


Fig. 10. Illustration of the origin of low-frequency fluctuations in the phase space projected on the $(N - N_{th}, \eta)$ plane and showing how attractor ruins of a compound-cavity mode merge with an antimode. The laser follows the trajectory shown on the top, moving from right to left and visiting many attractor ruins. After visiting the ruin around $\eta = -390$, the system moves back to the previous ruin, where it gets too close to the antimode (indicated by the arrow). The power drops out, and the instantaneous frequency (η/τ_{fb}) is reset to zero.

was later observed experimentally by Fischer et al. [106]. These observations are in good qualitative, and even quantitative, agreement with the predictions of the Lang–Kobayashi equations. Fig. 11 compares the experimental and theoretical

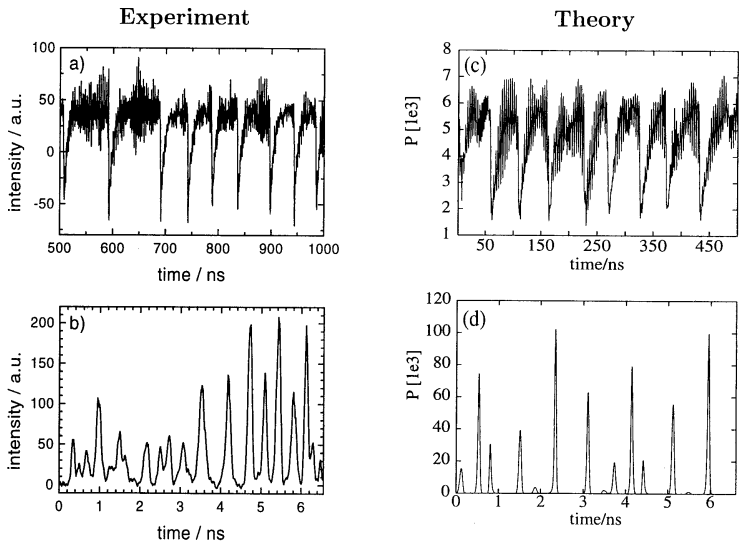


Fig. 11. Experimental (left column) and theoretical (right column) temporal variations of the output power of a semiconductor laser showing low-frequency fluctuations. The top row shows the filtered output while the bottom row shows the actual picosecond dynamics. (From Ref. [106], reprinted with permission).

results. It shows the filtered LFF signature in the top row together with the underlying *unfiltered* structure in the bottom row.

The mechanism of LFF has been termed *chaotic itinerancy with a drift* [106]. The itinerancy refers to the laser visiting a large number of the attractor ruins' of the compound-cavity modes. These ruins are clearly visible in Fig. 10 and are destabilized limit-cycles, bifurcated from the fixed points. The drift (from right to left in Fig. 10) refers to the trend of the laser to adjust its operating frequency to that corresponding to the minimum-threshold state, indicated by the arrow in Fig. 7. When operating in this state, the laser benefits maximally from the feedback. The reason for the frequency of the minimum threshold state to be different from the free running laser frequency is the α parameter. When $\alpha = 0$, the ellipse in Fig. 7 transforms into a circle. In that case, the minimum threshold state has the same frequency as the free running laser, and no LFF will occur. The stability properties of the minimum-threshold state have been discussed by Levine et al. [116].

The drop-outs are caused by the fact that, as the laser approaches the minimum threshold state, the distance between antimodes and attractor ruins decreases. Eventually the system will come too close to one of these antimodes, exhibiting a saddle–node instability, and a power drop-out occurs (indicated by the arrow in Fig. 10). Asymptotic expansion methods [115, 116] show that the fixed points close to this minimum-threshold point are the least unstable, and are, in fact, quite often stable. The basin of attraction of these points is so small that the system will not get there unless it is helped by some kind of large perturbation or controlling force [134, 135]. As the parameter space in which the phenomena of LFF and coherence collapse occur is enormous, regions exist where spontaneous-emission noise plays an important role in the actual manifestation of LFF [135, 136]. So far, no evidence has been found that other longitudinal modes (of the laser itself) are responsible for dynamic features.

3.4.6. Control of chaos

After the 1990 paper of Ott et al. discussed the possibility of controlling chaos [137], it did not take long before the first chaotic laser was controlled using a scheme known as the occasional proportional feedback [138]. The idea behind controlling chaos is as brilliant as it is simple. If one thinks of a chaotic attractor as a collection of unstable periodic orbits, the question arises whether it is possible to keep the system locked on one of those unstable orbits by applying a small control signal. In order to really 'control chaos' and not simply change the dynamical system, it is important to keep in mind that the control signal should vanish asymptotically with time.

Recent work on controlling the coherence collapse includes the use of a laser-diode interferometer [139–142], where chaos has been controlled with the delayed optoelectronic feedback using the technique of occasional proportional feedback. By changing the delay time, a variety of subharmonic periodic orbits can be stabilized with this technique. The synchronizing frequency can be readily

estimated from the ratio of the delay time and the system relaxation times. However, in the experiment of Ref. [139], the control signal did not vanish asymptotically with time. In later experiments [140, 142], theoretical analysis of the system was used to find the periodic orbit that would be stabilized with occasional proportional feedback. The technique of finding the unstable periodic orbits has been addressed theoretically in a number of recent papers [143–145].

3.5. Phase-conjugate feedback

Optical phase conjugation refers to a technique that converts an incoming signal of the form $E(r)\exp(-i\omega t)$ into $E^*(r)\exp(-i\omega t)$. In physical terms, phase conjugation reverses not only the direction of propagation but also the phase associated with each plane-wave component of an optical beam [146]. Many experimental techniques are available for realizing phase conjugation. Most notably among them are the ones involving the third-order nonlinear susceptibility $\chi^{(3)}$, that is responsible for nonlinear phenomena such as stimulated Brillouin scattering, stimulated Raman scattering, and four-wave mixing [146].

The most important application of phase conjugation is related to its distortion correction property: any static phase distortion occurring between the source and the phase conjugator is corrected automatically when the signal travels back to the source, since the phase-conjugated beam retraces exactly the path of the original beam. An interesting question, therefore, is how a semiconductor laser will react to the feedback from a phase-conjugate mirror. Most studies make use of a nonlinear medium pumped by two counter-propagating intense pump beams. The four-wave mixing process induced by $\chi^{(3)}$ leads to the emission of a phase-conjugate wave that propagates backward and may be more intense than the signal itself, because of the gain provided by the intense pump beams.

The first theoretical study of the effects of phase-conjugate feedback on the semiconductor-laser dynamics was published in 1991 by Agrawal and Klaus [147], who studied the degenerate four-wave mixing case in which frequencies of the four waves involved are identical. Van Tartwijk et al. [148] later extended this analysis to the case of nondegenerate four-wave mixing by allowing a frequency mismatch between the pump and signal beams. If we denote this frequency mismatch by $\delta \equiv \omega_s - \omega_p$, and modify Eq. (110) for the phase-conjugate feedback case, it can be written as [31]:

$$\frac{dA}{dt} = \frac{1}{2}(1 - i\alpha)G_N(N - N_{th})A + \gamma_{fb}A^*(t - \tau_{fb})\exp\left\{i\left[\left[2\delta\left(t - \frac{\tau_{fb}}{2}\right) + \varphi_{PCM}\right]\right]\right\}, \tag{122}$$

where γ_{fb} is the feedback-rate defined by Eq. (112), τ_{fb} is the round-trip time in the external cavity, and φ_{PCM} is a constant phase shift occurring at the phase-conjugate mirror. As a result of the nonlinear processes involved, the amplitude reflection coefficient of the phase-conjugate mirror may be greater than one, but it

is also a complex quantity [31]. The carrier-density equation is not affected by a change in the feedback mechanism and is still given by Eq. (111).

The effects of phase-conjugate feedback on the laser stability can be studied by using Eqs. (111) and (122), and following the same procedure as that used for the case of conventional optical feedback. By solving for the steady-state solutions, it is found that the phase-conjugate feedback case shows a remarkable similarity with the case of optical injection. Instead of locking its frequency to an external signal, the semiconductor now locks to the frequency of the pump laser used for four-wave mixing. The locking condition

$$\frac{|\delta|}{\gamma_{\mathrm{fb}}} \leq \sqrt{1+\alpha^2} \tag{123}$$

shows that the CW operation is possible only for relatively small detunings. This feature is similar to the locking range for optical injection.

Stability of the steady state is again examined by using the technique of linear-stability analysis. However, since the field equation, Eq. (122), involves a feedback delay that is absent in the injection-locking case, the stability of the CW state is governed by a different equation than that found in the case of injection locking. Stability boundaries have been numerically determined [148] and, as in the case of optical injection, three different regions can be identified: (i) stable locking, (ii) destabilized locking and (iii) non-locking. The dynamics outside the locking range show multi-wave mixing effects similar to those occurring in the injection case. Gray et al. [149] explored numerically the destabilized-locking region. It was found that, in general, semiconductor lasers with phase-conjugate feedback, display richer chaotic dynamics than those occurring in the case of conventional feedback. Period-doubling, quasi-periodicity, and intermittency routes to chaos, are observed in numerical simulations. Dramatic line narrowing (by locking to the pump beam) can occur for weak feedback conditions, a phenomenon predicted by Agrawal and Gray [150] using a stochastic analysis. Experimental verification of these predictions has been reported by several groups using a variety of phase-conjugation techniques [151–153]. Linewidth reduction from 5 MHz to 25 kHz by using phase-conjugate feedback has been demonstrated recently [154].

So far, the chaos induced by phase-conjugate feedback in semiconductor lasers has not been observed. An excellent candidate for such an experiment appears to be the setup of Ref. [154]. There, a broad-area semiconductor laser was used as the phase conjugator. The two counterpropagating beams inside the laser cavity act as pump beams in a four-wave-mixing scheme. Since the carrier lifetime in semiconductors is $\backsim$1 ns, the response time of the phase-conjugate mirror is also of the same order of magnitude [155–158]. Although relatively fast, such response is still sluggish for a comparison between theory and experiment, since almost all theoretical models assume an instantaneous response. Recently, it has been pointed out that the round-trip time within the nonlinear medium itself introduces sluggishness for the phase-conjugate feedback response even if the nonlinear medium responds instantaneously [159]. Using a wide-bandwidth phase-

conjugator, it has been predicted that by carefully choosing the detuning δ, mode-locking of different longitudinal modes can be achieved, resulting in the emission of short pulses [160]. Stable mode locking with pulse widths of $\backsim$30 ps and a timing jitter of less than 2 ps has been experimentally demonstrated using a $BaTiO_3$ crystal [161].

3.6. Spatial and polarization instabilities

Over the last 15 years or so, an enormous amount of research has been carried out to understand the nature of spatial instabilities, leading to pattern formation because of competition among various transverse modes of a laser [5, 43]. Topics of current interest include wide-aperture lasers, optical vortices, dough-nut modes, hexagon formation, and pattern recognition [5]. The mathematical treatment of spatial instabilities is quite cumbersome for most lasers since it requires inclusion of beam diffraction in both transverse dimensions x and y. As a result, even if the dispersive effects are neglected, Eq. (5) of Section 2 should be replaced with:

$$\frac{\partial A}{\partial z}+\frac{1}{v_0}\frac{\partial A}{\partial t}+\frac{1}{2i\beta_0}\left(\frac{\partial^2 A}{\partial x^2}+\frac{\partial^2 A}{\partial y^2}\right)=\frac{i\beta_0}{2\varepsilon_0 n_0^2}B, \tag{124}$$

where the slowly varying polarization B is still calculated by using the Bloch equations.

The problem is simplified considerably for semiconductor lasers in which diffraction needs to be included only along the lateral direction x, since the laser beam is confined along the other transverse dimension through dielectric waveguiding. The nonlinear dynamics of semiconductor lasers and amplifiers has been studied extensively in recent years [162]. Broad-area semiconductor lasers exhibit the phenomenon of beam filamentation at high powers, manifested through the break-up of the spatial beam profile into filaments and resulting in an ill-focusable output beam. It turns out that the filaments are not static, but change with time in a periodic or chaotic fashion depending on the pumping level. Considerable theoretical and experimental work has been carried out to understand the spatio-temporal nature of the filaments in broad-area semiconductor lasers [163, 164]. Fig. 12 shows an example of spatio-temporal dynamics in a broad-area laser, together with the prediction of numerical simulations based on the semiconductor Bloch equations [163]. A much simpler model, based on a two-level model for the gain, also shows spatio-temporal features that resemble qualitatively the features observed [165].

Recently, attempts have been made to identify the physical mechanisms responsible for filament formation and find ways to control the beam quality. It was shown theoretically [166] that several different nonlinear mechanisms can lead to filamentation. Spatial hole-burning and the index change associated with it through the α parameter play the most dominant role. Also important are the nonlinear phenomenon of self-focusing and self-defocusing occurring inside the active region at high pumping levels. In fact, under certain conditions, one may be

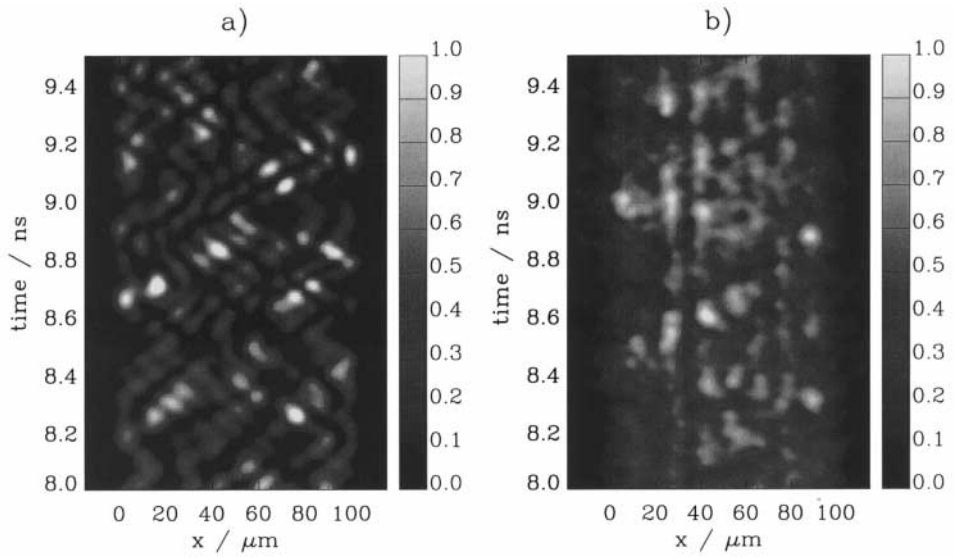


Fig. 12. Experimental (right) and theoretical (left) spatio-temporal dynamics of a broad-area semiconductor laser. (From Ref. [163], reprinted with permission).

able to use self-defocusing inside the specially designed cladding layers to counteract the α-induced spatial instabilities, resulting in a filamentation-free semiconductor laser [167]. The spatio-temporal nature of filamentation can be understood by performing a linear-stability analysis of the semiconductor-laser equations, modified to include the spatial effects such as carrier diffusion and beam diffraction. Analytic expressions for the oscillation frequency and the spacing between filaments have been obtained by using such a spatio-temporal linear-stability analysis [168]. Recently, numerical studies have investigated the possibility of controlling and suppressing these spatio-temporal instabilities using weak optical feedback from short cavities [169, 170].

Polarization instabilities refer to changes in the direction in which the optical field is polarized inside the laser cavity. In most semiconductor lasers, this direction remains unchanged, as the laser is pumped harder above its first threshold. An exception occurs in the case of vertical-cavity surface-emitting lasers (VCSELs). VCSELs are very 'hot' devices from the point of view of applications. They are promising candidates for applications such as optical data links and optical interconnects [57]. Owing to a relatively small cavity length ($\backsim 1\mu$m), VCSELs have such a large longitudinal-mode spacing ($\backsim 100$ THz), and only one longitudinal mode fits within the gain spectrum. As a result, VCSELs always operate in a single-longitudinal mode. However, VCSELs typically operate in several *transverse* modes. The nonlinear dynamics of these modes as a function of pump geometry, optical injection and optical feedback is of considerable interest [171–173]. Also, not all transverse modes have the same polarization, and polarization instabilities have been observed in these devices. A recently developed

model [174, 175] appears to agree very well with recent experiments [176, 177]. As mentioned earlier, this review does not attempt to cover spatial and polarization instabilities, and we refer to the literature for more details.

4. Fiber lasers

In this section, we turn to a new class of lasers, known as fiber lasers, that have been developed fully only over the last 10 years or so. These lasers use single-mode silica fibers as a host in which rare-earth ions are doped, which provide gain through optical pumping. The technology of doping silica fibers with rare-earth elements (lanthanides) was perfected in the late 1980 s. By using different dopants (such as erbium, neodymium and praseodymium), fiber lasers can be made to operate at various wavelengths. Erbium-doped fiber lasers have attracted the most attention, since they operate in the wavelength region near 1.55 μm that is of interest for lightwave communication systems. Such lasers are commonly pumped by using semiconductor lasers operating near 0.98 or 1.48 μm. This section deals with instabilities in fiber amplifiers and lasers. Fiber lasers differ from other lasers in at least one aspect: the gain medium is not the only nonlinear component within the cavity. The optical fiber itself has dispersive and nonlinear properties that need to be understood before one can attempt to tackle the problem of laser instabilities.

This section is organized as follows. We begin by considering a passive fiber (no gain) and show how Maxwell's wave equation leads to the nonlinear Schrödinger equation that governs wave propagation in optical fibers. This equation takes into account two important fiber properties known as the group-velocity dispersion (GVD) and self-phase modulation (SPM). We show that the combination of GVD and SPM can lead to an instability, called the *modulation instability*, even in the absence of gain and feedback. This instability is of an entirely different nature than those described earlier, as the perturbation growth requires propagation inside the optical fiber. The formation of optical solitons in fibers is intimately linked with modulation instability. We then focus on fiber amplifiers and show how modulation instability is modified by the addition of a gain medium. Finally, we address fiber lasers by including feedback within a laser cavity and focus on the link between modulation instability and Lorenz–Haken-type instabilities discussed in Sections 2 and 3.

4.1. Nonlinear Schrödinger equation

Electromagnetic fields propagating in optical fibers experience chromatic dispersion (GVD) and, if their intensity is large enough, SPM through nonlinear refraction. In Section 2 we neglected both of these effects by using a constant index of refraction n_0, resulting in the simple wave Eq. (5). We need to modify this equation to account for the GVD and SPM effects occurring inside optical

fibers. We follow the derivation given in Ref. [36], to which the reader is referred for more details.

We start with Maxwell's wave Eq. (1), except that we include all three space dimensions and split the polarization into linear and nonlinear parts such that:

$$\nabla^2\varepsilon - \frac{1}{c^2}\frac{\partial^2\varepsilon}{\partial t^2} = \mu_0\frac{\partial^2\mathcal{P}_{\mathrm{L}}}{\partial t^2} + \mu_0\frac{\partial^2\mathcal{P}_{\mathrm{NL}}}{\partial t^2}. \tag{125}$$

By writing the electric field and the polarization as:

$$\varepsilon(\mathbf{r}, t) = \frac{1}{2}E(\mathbf{r}, t)\exp(-i\omega_0 t) + \text{c.c.}, \tag{126}$$

$$\mathcal{P}_{\mathrm{j}}(\mathbf{r}, t) = \frac{1}{2}P_j(\mathbf{r}, t)\exp(-i\omega_0 t) + \text{c.c.}, \tag{127}$$

(with $j = L$ or NL) and assuming that the nonlinear response of silica fibers is instantaneous (a valid assumption in view of its electronic nature), the linear and nonlinear parts of the polarization can be written as:

$$P_{\mathrm{L}}(\mathbf{r}, t) = \varepsilon_0\int_{-\infty}^{+\infty}\chi_{xx}^{(1)}(\mathbf{r}, t - t')E(\mathbf{r}, t')\mathrm{d}t', \tag{128}$$

$$P_{\mathrm{NL}}(\mathbf{r}, t) = \varepsilon_0\varepsilon_{\mathrm{NL}}E(\mathbf{r}, t), \tag{129}$$

where $\chi_{xx}^{(1)}$ is the linear susceptibility, and the nonlinear contribution $\varepsilon_{\mathrm{NL}}$ resulting from the third-order susceptibility is given by:

$$\varepsilon_{\mathrm{NL}} = \frac{3}{4}\chi_{xxxx}^{(3)} \mid E(\mathbf{r}, t) \mid^2 . \tag{130}$$

The subscripts in $\chi_{xx}^{(1)}$ and $\chi_{xxxx}^{(3)}$ result from the assumption that the optical field maintains its polarization direction along the x-axis.

As long as $\varepsilon_{\mathrm{NL}}$ can be considered a small perturbation and assumed to vary slowly with time compared with ω_0^{-1}, we can ignore its time dependence while solving Eq. (125) in the frequency domain. The Fourier transform $\tilde{E}(\mathbf{r}, \omega)$ of the electric field $E(\mathbf{r}, t)$ satisfies the Helmholtz equation:

$$\nabla^2\tilde{E} + \varepsilon(\omega)k_0^2\tilde{E} = 0, \tag{131}$$

where $k_0 = \omega/c$ and

$$\varepsilon(\omega) = 1 + \tilde{\chi}_{xx}^{(1)}(\omega) + \varepsilon_{\mathrm{NL}}. \tag{132}$$

Eq. (131) can be solved by using the method of separation of variables and assuming a solution of the form:

$$\tilde{E}(\mathbf{r}, \omega) = F(x, y)\tilde{A}(z, \omega)\exp(i\beta_0 z), \tag{133}$$

where $F(x, y)$ denotes the spatial profile of the single-mode supported by the fiber.

Substituting Eq. (133) in Eq. (131), and using β^2 as the separation constant, we obtain:

$$\frac{\partial^2 F}{\partial x^2} + \frac{\partial^2 F}{\partial y^2} + [\varepsilon(\omega)k_0^2 - \tilde{\beta}^2]F = 0, \tag{134}$$

$$2i\beta_0 \frac{\partial \tilde{A}}{\partial z} + (\tilde{\beta}^2 - \beta_0^2)\tilde{A} = 0, \tag{135}$$

where we neglected the second derivative of $\tilde{A}$ with respect to z (SVEA).

One can solve Eqs. (134) and (135) by using the first-order perturbation theory, where the perturbation Δn is given by $\Delta n = [\varepsilon_{\mathrm{NL}} + \mathrm{Im}(\chi_{xx}^{(1)})]^{1/2}$. To first order, this perturbation does not affect the mode profile $F(x, y)$ obtained by solving Eq. (134) with $\Delta n = 0$. However, the eigenvalue $\beta(\omega)$ is affected by the perturbation Δn and becomes:

$$\tilde{\beta}(\omega) = \beta(\omega) + \Delta\beta, \tag{136}$$

where $\beta(\omega) = n(\omega)\omega/c$ with $n^2(\omega) = 1 + \mathrm{Re}(\chi_{xx}^{(1)})$ and

$$\Delta\beta \equiv \frac{k_0 \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \Delta n \mid F(x, y) \mid^2 \, \mathrm{d}x\,\mathrm{d}y}{\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mid F(x, y) \mid^2 \, \mathrm{d}x\,\mathrm{d}y}. \tag{137}$$

By expanding $\beta(\omega)$ in a Taylor expansion around ω_0, retaining terms up to the second order, and transforming the equation for $\tilde{A}$ back into the time domain, we obtain the following propagation equation for $A(z, t)$:

$$\frac{\partial A}{\partial z} + \frac{1}{v_g}\frac{\partial A}{\partial t} + \frac{i}{2}\beta_2 \frac{\partial^2 A}{\partial t^2} + \frac{\alpha_f}{2}A = i\gamma \mid A \mid^2 A, \tag{138}$$

where $v_g = (\mathrm{d}\beta/\mathrm{d}\omega)_{\omega = \omega_0}^{-1}$ is the group velocity at ω_0, and β_2 is the GVD parameter defined as:

$$\beta_2 = \left(\frac{\mathrm{d}^2\beta}{\mathrm{d}\omega^2}\right)_{\omega=\omega_0}. \tag{139}$$

The fiber loss α_f is related to $\mathrm{Im}(\chi_{xx}^{(1)})$, and the nonlinear parameter γ is defined as:

$$\gamma = \frac{3\omega_0 \chi_{xxxx}^{(3)}}{8 n_0 c a_{\mathrm{eff}}}, \tag{140}$$

where a_{eff} is the effective core area of the fiber given by

$$a_{\mathrm{eff}} \equiv \frac{(\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mid F(x, y) \mid^2 \, \mathrm{d}x\,\mathrm{d}y)^2}{\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mid F(x, y) \mid^4 \, \mathrm{d}x\,\mathrm{d}y}. \tag{141}$$

We have assumed that the field amplitude A is normalized such that $|A|^2$ represents optical power. The nonlinear coefficient γ is then expressed in units of $\mathrm{W}^{-1}/\mathrm{m}$.

Eq. (138) forms the central equation for studying wave propagation in optical fibers. It is based on three assumptions: (i) the SVEA, valid for pulse widths $\gg\omega_0^{-1}$, (ii) the instantaneous nonlinear response of silica fibers, and (iii) the neglect of higher-order nonlinear and dispersive effects (valid for pulse widths larger than a few picoseconds). See Ref. [36] for further details.

Before solving Eq. (138), we transform it to a reference frame moving along with the pulse at its group velocity (known as the retarded frame) by using

$$z' = z, \quad T = t - z/v_g, \tag{142}$$

and obtain

$$\frac{\partial A}{\partial z'} = -\frac{1}{2}\alpha_f A - \frac{i}{2}\beta_2 \frac{\partial^2 A}{\partial T^2} + i\gamma \mid A \mid^2 A. \tag{143}$$

The fiber loss α_f is quite small in silica fibers, and is often compensated through periodic amplification of the signal if the loss becomes too large. Therefore, we neglect the fiber loss initially but will include it when we consider fiber lasers. We also drop the prime in z' for notational simplicity. Eq. (143) then becomes:

$$i\frac{\partial A}{\partial z} = \frac{1}{2}\beta_2 \frac{\partial^2 A}{\partial T^2} - \gamma \mid A \mid^2 A. \tag{144}$$

This equation is known as the nonlinear Schrödinger equation. It not only has CW solutions but also provides interesting pulse-like solutions known as solitons. The nonlinear Schrödinger equation is encountered in many different fields, including nonlinear optics [36] and optical communications [178]. In the next section we find the steady-state solution of this equation and study its stability.

4.1.1. Modulation instability in passive fibers

The steady-state solution of the nonlinear Schrödinger equation Eq. (144) is obtained by setting $\partial^2 A/\partial T^2 = 0$, and is given by:

$$A_s(z) = \sqrt{P_0}\exp(i\gamma P_0 z), \tag{145}$$

where P_0 is the optical power of a CW beam incident at the input end of the fiber located at $z = 0$. The power of the CW beam remains unchanged upon propagation, but it acquires a nonlinear phase shift $\varphi_{NL} = \gamma P_0 z$. From our perspective, the relevant question is whether the CW solution in Eq. (145) is stable against small perturbations that invariably occur in practice.

The linear-stability analysis used in previous sections can be used to address the stability issue, except for a major difference that the perturbation should be allowed to vary both in space and time. By substituting the perturbed steady state:

$$A(z, T) = [1 + u(z, T) + iv(z, T)]A_s(z), \tag{146}$$

in Eq. (144) and linearizing it with respect to small perturbations u and v, we obtain the following set of two coupled but linear equations:

$$\frac{\partial v}{\partial z} = 2\gamma P_0 u - \frac{1}{2}\beta_2 \frac{\partial^2 u}{\partial T^2}, \tag{147}$$

$$\frac{\partial u}{\partial z} = \frac{1}{2}\beta_2 \frac{\partial^2 v}{\partial T^2}. \tag{148}$$

These linearized equations can be easily solved by assuming that the perturbations propagate in the form of plane waves, i.e.

$$u(z, T) = u_0 \exp[i(Kz - \Omega T)], \quad v(z, T) = v_0 \exp[i(Kz - \Omega T)]. \tag{149}$$

Here, Ω is real and denotes the frequency of the perturbation, while K is the propagation constant of the plane wave and is allowed to be complex. In fact, when the imaginary part of K is negative, perturbations at frequency Ω will grow upon propagation, as is evident from Eq. (149). This situation is similar to that occurring for a positive value of s in Eq. (48). The underlying steady state is unstable in both cases, although the growth of perturbations follows quite different patterns. In the case of modulation instability, perturbations are in the form of a plane wave whose amplitude increases with propagation. In contrast, for a Lorenz–Haken-type instability perturbations grow in time while remaining spatially unaffected.

To see under what conditions K can become complex, we substitute Eq. (149) in Eqs. (147) and (148), and obtain the following matrix equation:

$$\begin{pmatrix} 2iK & \beta_2\Omega^2 \\ \beta_2\Omega^2 + 4\gamma P_0 & -2iK \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = 0. \tag{150}$$

Since a nontrivial solution of Eq. (150) exists only when the determinant of the coefficient matrix is zero, we obtain the following dispersion relation expressing the relation between the perturbation frequency Ω and the associated propagation constant K:

$$K = \pm\frac{1}{2} \mid \beta_2 \mid \Omega\sqrt{\Omega^2 + \mathrm{sgn}(\beta_2)\Omega_c^2}, \tag{151}$$

where $\mathrm{sgn}(\beta_2) = +1$ or -1 depending on whether GVD is normal or anomalous, and the critical frequency Ω_c is given by:

$$\Omega_c = \sqrt{\frac{4\gamma P_0}{\mid \beta_2 \mid}}. \tag{152}$$

It is obvious from Eq. (151) that, only when the GVD is anomalous, there is a possibility for perturbations to grow since K will then have a negative imaginary part for frequencies such that $\Omega < |\Omega_c|$. The gain spectrum of modulation instability is thus given by:

$$g(\Omega) \equiv -2\,\mathrm{Im}[K(\Omega)] = \mid \beta_2\Omega \mid \sqrt{\Omega_c^2 - \Omega^2}. \tag{153}$$

Fig. 13 plots Eq. (153) by choosing $\beta_2 = -20$ ps^2/km and $\gamma = 2$ W^{-1}/km, typical values for silica fibers in the wavelength region near 1.5 μm. For a CW beam launched with 4 W of input power, the gain exists for perturbation frequencies in the range 0–0.2 THz. Physically, modulation instability can be understood in terms of a four-wave-mixing process that is phase-matched through self-phase modulation [36]. In that picture, two photons from the CW beam at frequency ω_0 are converted into two photons, one at $\omega_0 + \Omega$ and one at $\omega_0 - \Omega$ when $|\Omega| < \Omega_c$.

4.1.2. Optical solitons

As is the case with any linear analysis of a nonlinear problem, the exponential growth of the perturbations can not go on indefinitely. As a result of the growing perturbations, the CW beam becomes depleted, resulting in changes in the gain and the frequency range of modulation instability. How the CW beam evolves beyond the validity regime of a linear-stability analysis, can be found only by solving the nonlinear Schrödinger equation directly. Numerical simulations show [179] that a CW beam, weakly perturbed by a sinusoidal modulation, evolves into a train of narrow pulses separated by the period of the initial modulation. When the initial modulation is broad band and contains many frequencies, the narrow pulses are separated by the period corresponding to the frequency $\Omega_{max} = (2\gamma P_0/|\beta_2|)^{1/2}$ for which modulation instability gain is maximum.

Since the modulation instability only occurs in the anomalous-dispersion regime of optical fibers, it is clear that the nonlinear Schrödinger equation has a fundamentally different character for positive and negative values of β_2. In the case of anomalous dispersion ($\beta_2 < 0$), an optical pulse can propagate through the fiber without changing its shape under appropriate conditions. Such pulses result from a balance between the effects of SPM and GVD and are called *optical solitons*.

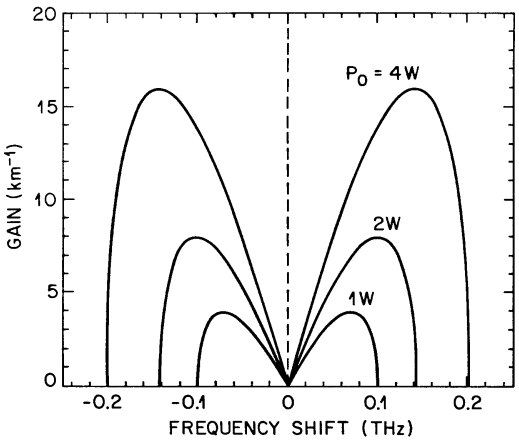


Fig. 13. Gain spectra of modulation instability at three power levels. Fiber parameters are $\beta_2 = -20$ ps^2/km and $\gamma = 2$ W^{-1}/km.

Since SPM and GVD play such crucial roles, it is useful to identify two length scales, the dispersion length L_D and the nonlinear length L_{NL}, for a pulse with power P_0 and characteristic width T_0. These lengths are defined as

$$L_D = T_0^2/|\beta_2|, \quad L_{NL} = 1/\gamma P_0. \tag{154}$$

We can normalize the nonlinear Schrödinger equation, Eq. (144) in terms of the so-called soliton units by using:

$$u = \frac{NA}{\sqrt{P_0}}, \quad \xi = \frac{z}{L_D}, \quad \tau = \frac{T}{T_0}, \quad N^2 = \frac{L_D}{L_{NL}}, \tag{155}$$

and write it in its standard form [assuming $\mathrm{sgn}(\beta_2) = -1$]:

$$i\frac{\partial u}{\partial \xi} + \frac{1}{2}\frac{\partial^2 u}{\partial \tau^2} + |u|^2 u = 0. \tag{156}$$

By employing the inverse-scattering method [180], the soliton solutions of Eq. (156) are found to exist only when N is an integer. The fundamental soliton ($N = 1$) is found to be:

$$u(\xi, \tau) = \mathrm{sech}(\tau)\exp(i\xi/2). \tag{157}$$

This result implies that if a hyperbolic-secant-shaped optical pulse, whose peak power P_0 and pulse width T_0 are chosen such that $N = 1$, is launched into an ideal lossless fiber, the pulse will propagate without any change for arbitrarily long distances. From Eq. (157) it is readily seen that the full width of the fundamental soliton at half-maximum point is $2\ln(1+\sqrt{2})T_0 \simeq 1.76\,T_0$. For dispersion-shifted fibers with $\beta_2 = -2$ ps^2 km^{-1} and $\gamma = 4$ W^{-1} km^{-1}, a fundamental soliton with $T_0 = 10$ ps requires a peak power of only 5 mW, power levels readily available from semiconductor lasers. The higher-order solitons of the same width require a launching power that is N^2 times larger. Their initial shape is given by:

$$u(0, \tau) = N\,\mathrm{sech}(\tau). \tag{158}$$

All higher-order solitons follow a periodic evolution pattern with period $z_0 = \pi L_D/2$, i.e. the pulse shape of these solitons changes with propagation but recovers its original shape every integer multiple of z_0.

A natural question to ask is: how stable are optical solitons? what happens if the input pulse at $\xi = 0$ does not have a hyperbolic-secant shape or has a power such that N is not an integer? Other questions concern the effects of fiber loss, higher-order dispersion, and higher-order nonlinear effects. Based on the inverse-scattering method, a perturbation theory of solitons has been developed that can answer most of these questions. Since this perturbation theory is quite involved and falls outside the scope of this review, we refer only to recent literature [178].

4.2. Fiber amplifiers

To overcome the fiber loss in long-haul optical communication systems, the signal propagating inside the fiber needs to be amplified periodically. Typical distances between amplifiers are in the range of 50–100 km. Fiber amplifiers are commonly used for this purpose. For 1.55 μm lightwave systems, the use of the erbium-doped fiber amplifiers has become the standard. In this section we discuss how modulation instability is affected by the dopants that provide gain in fiber amplifiers.

4.2.1. Ginzburg–Landau equation

Before considering the Bloch model of rare-earth dopants, we set up an equation that has a well-defined mathematical relation with the nonlinear Schrödinger equation. This equation belongs to the class of *complex* Ginzburg–Landau equations and has specific solitary-wave solutions, sometimes referred to as autosolitons.

The effects of dopants on an optical pulse propagating inside a fiber amplifier can be included by adding a term $\mu_0(\partial^2 P_d/\partial t^2)$ to the right side of Eq. (125), where P_d is the dopant-induced polarization. We can then follow the analysis of Section 2.1 and introduce the slowly-varying polarization B through Eq. (4). The derivation given in Section 4.1 can still be carried out to derive Eq. (144), but a term containing B should be added to it, resulting in:

$$i\frac{\partial A}{\partial z} = \frac{1}{2}\beta_2 \frac{\partial^2 A}{\partial T^2} - \gamma \mid A \mid^2 A - \frac{1}{2}B, \tag{159}$$

where B is obtained by solving the Bloch equations of Section 2, Eqs. (28) and (29).

The dynamics of dopants modeled as two-level atoms is governed by the population decay time T_1 and the dipole-dephasing time T_2. When the assumption is made that T_1 is much larger than all other time scales of interest, and that the gain remains unsaturated and can be replaced with its small-signal value g_0, Eq. (28) can be solved in the Fourier domain, resulting in the following analytic expression for the dopant susceptibility:

$$\tilde{\chi}(\omega) = \frac{\tilde{B}}{\beta_0 \tilde{A}} = \frac{g_0/\beta_0}{(\omega - \omega_A)T_2 + i}, \tag{160}$$

where ω_A is the dopant-resonance frequency. Since the dipole-dephasing time T_2 is relatively small for doped fibers ($\backsim$100 fs), one often approximates the dopant susceptibility by its Taylor expansion up to second order. Physically, this approximation implies that the Lorentzian gain profile is approximated by a parabola. Transforming $\tilde{B}$ from Eq. (160) back into the time domain, it turns out that the GVD parameters β_2 are modified by the dopants as [181]:

$$\beta_2^{\text{eff}} = \beta_2 + i\frac{g_0 T_2^2}{(1 - i\delta)^3}. \tag{161}$$

The group velocity is also affected by dopants but the change is relatively small and can be neglected. In the resonant case ($\delta = 0$), the field-propagation Eq. (159) for optical amplifiers becomes:

$$\frac{\partial A}{\partial z} = \frac{1}{2} g_0 A + \frac{1}{2}(b - i\beta_2)\frac{\partial^2 A}{\partial T^2} + i\gamma \mid A \mid^2 A. \tag{162}$$

where $b = g_0 T_2^2$ is the dopant-induced correction to the GVD. Eq. (162) is a complex Ginzburg–Landau equation. A similar equation has been studied extensively in the field of fluid dynamics [182].

4.2.2. Modulation instability in fiber amplifiers

To study how modulation instability is affected by the dopant-induced gain in fiber amplifiers, we analyze the stability of the CW solution of the Ginzburg–Landau equation [183] For a CW input beam with power P_0, the steady-state solution of Eq. (162) is easily found to be:

$$A_s(z) = \sqrt{P_0}\exp\left(i\gamma P_0 \int_0^z \exp(g_0 z)\,\mathrm{d}z + \frac{g_0 z}{2}\right), \tag{163}$$

and is a generalization of Eq. (145) obtained for the undoped fiber. When the steady state is subjected to a small perturbation having a functional form similar to Eq. (149), we can easily obtain the linearized equations for the perturbations u and v. By assuming a plane-wave type solution for u and v of the form

$$\eta(z, t) = \eta_0(z, t)\exp(i\int K(z)\,\mathrm{d}z - i\Omega t) \quad (\eta = u, v), \tag{164}$$

the dispersion relation that links the frequency of the perturbation with its local wavenumber $K(z)$ is found to be:

$$K(\Omega, z) = \frac{i}{2} g_0 T_2^2 \Omega^2 \pm \mid \beta_2 \mid \Omega[\Omega^2 + \mathrm{sgn}(\beta_2)\Omega_c^2 \exp(g_0 z)]^{1/2}, \tag{165}$$

where Ω_c is given by Eq. (152). Modulation instability occurs whenever K has a negative imaginary part because perturbations u and v then grow at a faster rate than the CW beam. It is useful to define the total integrated gain for the modulation instability at a frequency Ω as

$$h(\Omega) = -2\int_0^L \mathrm{Im}[K(\Omega, z)]\,\mathrm{d}z. \tag{166}$$

In the normal-dispersion regime, $\mathrm{sgn}(\beta_2) = +1$, $h(\Omega)$ is negative for all frequencies, indicating that any perturbation will amplify less than the signal. In the anomalous-dispersion regime, this is not the case. As is shown in Fig. 14, the threshold for modulation instability is substantially reduced compared with the

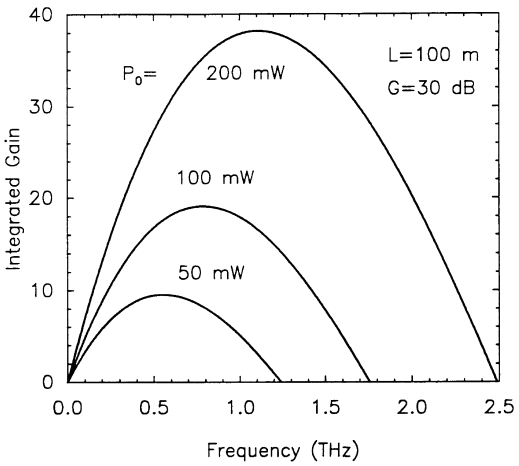


Fig. 14. Gain spectra of modulation instability at several input levels for a 100 m long fiber amplifier with 30 dB gain obtained by using the Ginzburg–Landau model.

passive-fiber case. A 100 mW input beam will readily show modulation instability and break up in a pulse train at a repetition rate $\backsim$1 THz in a 100 m long EDFA operating with 30 dB unsaturated gain [183].

4.2.3. Maxwell–Bloch model for modulation instability

The analysis of the previous section is not totally accurate because it approximates the gain spectrum of a fiber amplifier with a parabola, resulting in the Ginzburg–Landau Eq. (162). In this section, we investigate the occurrence of modulation instability in a doped fiber without this restriction. For this purpose, we supplement the field Eq. (159) with the Bloch Eqs. (28) and (29), and obtain the following set of Maxwell–Bloch equations:

$$\frac{\partial A}{\partial z} = \frac{i}{2}B - \frac{i\beta_2}{2}\frac{\partial^2 A}{\partial T^2} + i\gamma \mid A \mid^2 A, \tag{167}$$

$$T_2\frac{\mathrm{d}B}{\mathrm{d}T} = -(1 - i\delta)B - iAg, \tag{168}$$

$$T_1\frac{\mathrm{d}g}{\mathrm{d}T} = g_0 - g + \frac{\mathrm{Im}(A^*B)}{P_{\mathrm{sat}}}, \tag{169}$$

where A has units such that $|A|^2$ represents optical power, rather than intensity, and

$$P_{\mathrm{sat}} = I_{\mathrm{sat}}a_{\mathrm{eff}} \tag{170}$$

is the saturation power of the dopants and is related to I_{sat} of Section 2 through the effective core area given by Eq. (41).

We proceed by looking for the steady-state solution of the Maxwell–Bloch equations, Eqs. (167)–(169). These equations are valid for a wide range of values for T_1 and T_2, and the gain saturation is fully taken into account. The steady-state solution for a CW beam launched with power P_0 at $z = 0$ is readily obtained and is given by [184]:

$$A_s(z) = \sqrt{P_A(z)}\exp[i\varphi_s(z)], \tag{171}$$

$$B_s(z) = \frac{1}{\delta + i} A_s(z) g_s(z), \tag{172}$$

$$g_s(z) = g_0\left[1 + \frac{P_A(z)/P_{sat}}{1+\delta^2}\right]^{-1}. \tag{173}$$

Using the imaginary part of Eq. (167), the phase profile $\varphi_s(z)$ can be written in terms of the power profile $P_A(z)$ as:

$$\varphi_s(z) = \gamma \int_0^z P_A(z')\,dz' + \frac{\delta/2}{1+\delta^2}\int_0^z g_s(z')\,dz'. \tag{174}$$

From the real part of Eq. (167) and from Eq. (173), the normalized optical power, defined as $f(z) = P_A(z)/[P_{sat}(1 + \delta^2)]$, is found to satisfy

$$\frac{df}{dz} = \frac{g_0}{1+\delta^2}\frac{f}{f+1}. \tag{175}$$

This differential equation can be solved easily and results in the following transcendental equation for $P_A(z)$:

$$\ln\left(\frac{P_A(z)}{P_0}\right) + \frac{P_A(z) - P_0}{P_{sat}} = \frac{g_0 z}{1+\delta^2}. \tag{176}$$

After the power profile $P_A(z)$ is found, the gain $g_s(z)$, the polarization $B_s(z)$, and the phase $\varphi_s(z)$ follow from Eqs. (171)–(174).

We next perform a linear-stability analysis of the CW solution by using the Maxwell–Bloch equations, Eqs. (167)–(169). Considering small perturbations in a form similar to those in Eq. (149), linearizing the five resulting equations, and using Eq. (164), the following dispersion relation is obtained at resonance ($\delta = 0$):

$$\begin{aligned}
&\{[2iK(z) + g_s(z)]^2 + \beta_2^2\Omega^2[\Omega^2 + \mathrm{sgn}(\beta_2)\Omega_c^2(z)]\}\\
&\quad\times(1 - i\Omega T_2)[(1 - i\Omega T_1)(1 - i\Omega T_2) + I(z)]\\
&\quad - g_s(z)[2iK(z) + g_s(z)](1 - i\Omega T_2)[1 - i\Omega T_1 - I(z)]\\
&\quad - g_s(z)[2iK(z) + g_s(z)][(1 - i\Omega T_1)(1 - i\Omega T_2) + I(z)]\\
&\quad + g_s^2(z)[1 - i\Omega T_1 - I(z)] = 0,
\end{aligned} \tag{177}$$

where $I(z) = P_A(z)/P_{sat}$, and $\Omega_c(z) = (4\gamma P_A/|\beta_2|)^{1/2}$ is the z-dependent critical frequency. Before we examine the implications of Eq. (177), we note that the Rabi

frequency, a well-known quantity for two-level atoms [8], can be written as:

$$\Omega_{\mathrm{Rabi}}(z) = \left| \frac{\mu A_{\mathrm{s}}(z)}{\hbar} \right| = \sqrt{\frac{P_{\mathrm{A}}(z)}{P_{\mathrm{sat}} T_1 T_2}}. \tag{178}$$

The dispersion relation, Eq. (177), depends on a large number of parameters. We focus on the effects of the dipole-dephasing time T_2. For most fiber amplifiers, T_2 is $\backsim$100 fs, resulting in a wide gain spectrum. However, by cooling the fiber, the dephasing processes can be slowed down substantially, and values of $T_2 \backsim 10$ ps becomes possible [185]. In Fig. 15 we show the modulation-instability gain spectrum for several values of T_2 in the range of 0.1–10 ps for a fiber amplifier with a 30 dB small-signal gain, i.e. $\exp(g_0 L) = 1000$. All other parameters are given in the caption. Note that the saturation power P_{sat} is inversely proportional to T_2 through Eq. (170). By satisfying Eq. (170) for each value of T_2, the Rabi frequency is kept constant for all curves at a value $\Omega_{\mathrm{Rabi}} = 1.29 \times 10^{-3}\ \Omega_{\mathrm{c}}$. By increasing the dephasing time T_2, the two following trends are observed.

Firstly, as seen in Fig. 15, an increase in T_2 shrinks the bandwidth and reduces the peak gain for modulation instability. When $T_2 = 100$ fs ($P_{\mathrm{A}}/P_{\mathrm{sat}} = 0.01$), modulation instability occurs for frequencies up to $\Omega_{\mathrm{c}}/2\pi = 3.9$ GHz, and the peak gain is found at $\Omega = \Omega_{\mathrm{c}}/\sqrt{2}$. The peak gain is very close to the analytical value $2\gamma P_{\mathrm{A}}$, the value obtained in the Ginzburg–Landau limit [183]. Already when $T_2 = 2$ ps ($P_{\mathrm{A}}/P_{\mathrm{sat}} = 0.2$), the gain bandwidth has shrunken by 40%. Around $T_2 = 8$ ps ($P_{\mathrm{A}}/P_{\mathrm{sat}} = 0.8$) modulation instability has almost ceased to occur. For $T_2 = 21$ ps ($P_{\mathrm{A}}/P_{\mathrm{sat}} = 2.1$), the modulation instability vanishes completely. Long before that happens, the gain becomes so small that it is doubtful that modulation instability can be observed in a single-pass amplifier.

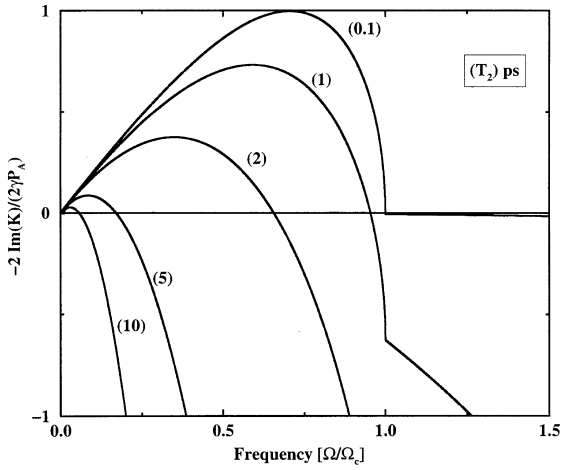


Fig. 15. Gain spectra of modulation instability for severals values of T_2 using the Maxwell–Bloch equations for a fiber amplifier. Parameter values used are: $g_0 L = 6.91$, $P_0 = 1$ mW, $T_1 = 0.1$ ms, $\beta_2 L = -20$ ps^2/, $\gamma L = 3$ W^{-1}, and $P_{\mathrm{sat}} = 1$ mW when $T_2 = 0.1$ ps.

Secondly, as the gain spectrum shrinks with increasing T_2, a second peak begins to develop, an effect caused directly by the dopants since it involves frequencies close to Ω_{Rabi} (see Fig. 16). Around the Rabi frequency, a secondary gain peak begins to form for $T_2 > 20$ ps; the peak value becomes positive around $T_2 = 80$ ps and increases with T_2. When T_2 is increased further, the gain spectrum slowly returns to its original width and strength (out of scale in Fig. 16). Around $T_2 = 11.5$ ns ($P_A/P_{\text{sat}} = 1150$), the gain spectrum shows again a peak developing in the vicinity of Ω_c. At the highly improbable value of $T_2 \backsim 1$ μs ($P_A/P_{\text{sat}} = 10^5$), the modulation instability spectrum becomes quite similar to that occurring for $T_2 = 100$ fs.

Thus, we find four regimes of dopant dynamics depending on the value of the dipole-dephasing time T_2. In the first regime (0.1 ps $< T_2 <$ 21 ps), increasing T_2 leads to a reduced range and lower gain for the modulation instability. In the second regime (21 $< T_2 <$ 80 ps), no modulation instability occurs, and a new peak around the Rabi frequency begins to form. In the third regime (80 ps $< T_2 <$ 11.5 ns), the gain of the new peak becomes positive, and modulation instability occurs around the Rabi frequency. In the fourth regime corresponding to the long T_2 limit ($T_2 > 1$ μs), the gain spectrum recovers fully to its original form. The boundaries between these regimes depend, of course, on the power level P_A. We note that the gain peak around Ω_c is insensitive to changes in T_1 as long as such changes are accompanied by a change in the saturation power P_{sat}. However, if we keep the saturation power constant upon changing T_1 (this can be done numerically by adjusting the dipole moment μ), a decrease in T_1 leads to stabilization of the system and a reduction in the instability gain.

We emphasize that the narrow gain peak around Ω_{Rabi} is so weak that it is questionable whether its effects can be observed in an amplifier. In the case of a

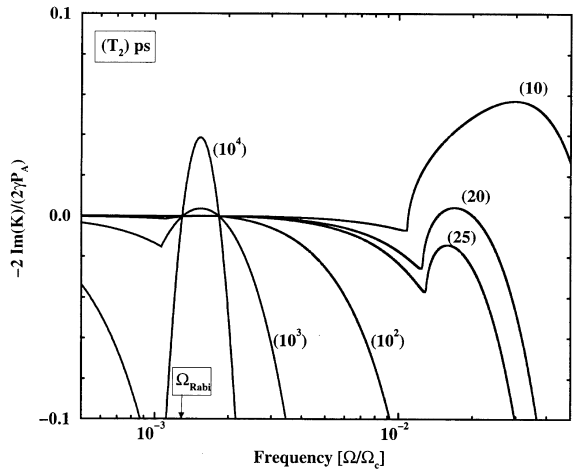


Fig. 16. Same as in Fig. 15 except for larger values of T_2. In the range 21 $< T_2 <$ 80 ps, modulation instability is totally quenched. When T_2 exceeds 100 ps, a narrow gain peak forms around the Rabi frequency.

laser, however, such a weak gain may build-up to a substantial level over many round trips. It should also be noted that this gain peak does not depend on the sign of β_2 and occurs in both the normal- and anomalous-dispersion regimes. Apart from this new (and for realistic systems extremely weak) instability, the Maxwell–Bloch model agrees with the Ginzburg–Landau model rather well. Of course, the quantitative differences become larger as the approximations leading to the Ginzburg–Landau model become inappropriate.

4.3. Fiber lasers

Although the first fiber laser was demonstrated as early as 1962 [186], it was only during the early 1990 s that the development of fiber lasers matured. Much attention has focused on erbium-doped fiber lasers, since such lasers are capable of producing ultrashort pulses in the 1.55 μm wavelength region through active or passive mode locking. Several books discuss the design and operating characteristics of fiber lasers [36, 187–189].

In its most simple configuration, a fiber laser is nothing but an amplifier whose two ends are connected through a fiber coupler to form a ring cavity. The feedback provided by the laser converts an optical amplifier into a laser oscillator. Similar to the case of Section 2, the Maxwell–Bloch equations obtained for a fiber amplifier, Eqs. (167)–(169), can still be used to study the laser dynamics, but one must impose the appropriate boundary conditions.

The central question, therefore, is: what happens to modulation instability when the optical field experiences feedback in a laser cavity? The answer lies in the longitudinal modes supported by a laser cavity. As a plane-wave-type perturbation grows exponentially with the gain provided by the modulation instability, it must eventually satisfy the boundary conditions. As a result, the intracavity spatial distribution of the perturbation must match with that associated with one or more longitudinal modes of the cavity. This requirements affects the range of possible values of the propagation constant K associated with the perturbation. Without the boundary conditions, modulation instability will cause the plane-wave component with the largest instability gain to grow. The feedback in the cavity changes the situation dramatically by imposing that only values of K associated with the longitudinal modes are possible.

It turns out that from a nonlinear-dynamics point of view, one should distinguish between the *convective* and *absolute* instabilities [190]. Convective instabilities exist by virtue of propagation, while absolute instabilities are purely temporal. The Lorenz–Haken instabilities of Section 2 and the semiconductor-laser instabilities of Section 3 can be classified as absolute, since the laser was assumed to operate in a single longitudinal mode in both cases. The relevant time scale of the laser dynamics in those cases is, by definition, longer than the round-trip time in the laser cavity. In the case of fiber lasers, the optical field evolves considerably on a time scale shorter than the round-trip time because of the GVD and SPM effects occurring within the cavity. As a result, one must distinguish

between the laser dynamics occurring over these two different time scales before one can understand how a convective-type modulation instability evolves into a Lorenz–Haken-type instability in fiber lasers. We discuss in this section both types of instabilities and show how the Lorenz–Haken equations of Section 2 can be generalized to include the effects of GVD and SPM occurring invariably in fiber lasers.

4.3.1. Modulation instability in fiber lasers

To understand how modulation instability is affected by the feedback in the laser cavity, we modify the Maxwell–Bloch equations obtained for fiber amplifiers, Eqs. (167)–(169), to account for the cavity losses. The main difference consists of adding a loss term to Eq. (167), resulting in:

$$\frac{\partial A}{\partial z} = \frac{i}{2}B - \frac{\alpha_c}{2}A - \frac{i\beta_2}{2}\frac{\partial^2 A}{\partial T^2} + (\theta + i\gamma) \mid A \mid^2 A, \tag{179}$$

where the cavity loss α_c now also includes all sources of losses (fibers, mirrors, output coupler, etc.). We have added another nonlinear term $\theta|A|^2A$ on the right side of Eq. (179) to account for intensity-dependent losses within the laser cavity. For positive values of θ, this term can account for the effects of a fast-saturable absorber, often introduced inside the laser cavity to induce passive mode locking [36]. For negative values of θ, this term accounts for the nonlinear phenomenon of two-photon absorption. The Bloch equations remain unchanged and are the same as those used for fiber amplifiers.

By performing a linear-stability analysis of the modified Maxwell–Bloch equations, similar to the case of fiber amplifiers, we can find the initial growth of a localized perturbation through a convective-type modulation instability. However, such a convective growth does not necessarily imply that the perturbation would survive after many round trips. To ensure that, one must numerically solve the Maxwell–Bloch equations and impose the boundary conditions at the mirrors.

The main difference between the modulation-instability analysis of fiber amplifiers and fiber lasers is that we can assume a *uniform* intensity profile along the cavity for the CW solution of the Maxwell–Bloch equations if cavity losses are relatively small. The CW solution of Eqs. (167)–(169) is then given by:

$$A_s(z) = \sqrt{P_0}\exp[i\varphi_s(z)], \tag{180}$$

$$B_s(z) = \frac{1}{\delta + i}A_s g_s, \tag{181}$$

$$g_s = g_0\left[1 + \frac{P_0/P_{sat}}{1+\delta^2}\right]^{-1}, \tag{182}$$

where the laser power P_0 and the phase profile $\varphi_s(z)$ are obtained from:

$$g_s = (\alpha_c - 2\theta P_0)(1 + \delta^2), \tag{183}$$

$$\frac{d\varphi_s}{dz} = \gamma P_0 + \frac{\delta g_s}{2(1 + \delta^2)}. \tag{184}$$

Since Eq. (183) is quadratic in P_0 [see Eq. (182)], in principle, two values for the laser power are found; the larger one is unphysical and corresponds to an unstable solution [191].

A straightforward linear-stability analysis of the above CW solution leads to the following dispersion relation at resonance ($\delta = 0$)

$$\begin{aligned}
&\{[2iK + \alpha_c - 6\theta P_0][2iK + \alpha_c - 2\theta P_0] + \beta_2^2\Omega^2[\Omega^2 + \mathrm{sgn}(\beta_2)\Omega_c^2]\} \\
&\quad \times (1 - i\Omega T_2)[(1 - i\Omega T_1)(1 - i\Omega T_2) + I_0] \\
&\quad - gs[2iK + \alpha_c - 2\theta P_0](1 - i\Omega T_2)[1 - i\Omega T_1 - I_0] \\
&\quad - gs[2iK + \alpha_c - 6\theta P_0][(1 - i\Omega T_1)(1 - i\Omega T_2) + I_0] \\
&\quad + g_s^2[1 - i\Omega T_1 - I_0] = 0.
\end{aligned} \tag{185}$$

Here, $I_0 = P_0/P_{sat}$ and $\Omega_c = (4\gamma P_0/|\beta_2|)^{1/2}$ is the critical frequency. This dispersion relation is identical to Eq. (177) when one replaces K, P_0, g_s by their z-dependent counterparts and uses Eq. (183).

The dispersion relation, Eq. (185), reduces to the one previously obtained by Chen et al. [191] in the Ginzburg–Landau limit. In contrast with the amplifier case described in the previous section, lasers generally operate in the heavily saturated regime. This means that the instability around the Rabi frequency is now more likely to play a significant role.

We first compare the predictions of Eq. (185) with the Ginzburg–Landau model, using the same parameters as in Ref. [191]. In Fig. 17 we show the differences for a figure-eight laser [36]. Although the trajectories of K as a function of frequency Ω are quite different in the two cases, the resulting gain spectra agree quite well, at least in the central region. The frequency range over which the gain occurs is underestimated by about 10% by the Ginzburg–Landau model. Both models show vanishing gain near 100 kHz, as indicated by the vertical line at $\Omega \backsim 0$ in Fig. 17(a). The Rabi frequency in this case is $\Omega_{Rabi} = 55$ MHz, while Ω_c is larger by almost three orders of magnitude ($\Omega_c = 37$ GHz). Even so, modulation instability occurs for frequencies almost twice as large as Ω_c.

The results for dye-laser parameters are shown in Fig. 18. The Maxwell–Bloch model predicts that modulation instability occurs in a narrow band around 30 GHz, whereas the Ginzburg–Landau model predicts no instability at all! The Rabi frequency Ω_{Rabi} and the critical frequency Ω_c are both close to 24 GHz, indicating strong interaction between the fiber-host and two-level nonlinearities. This case is an example in which significant qualitative differences are found to occur between the predictions of the Maxwell–Bloch equations and the Ginzburg–Landau equation. Not surprisingly, the laser power in Fig. 18 is about 60 times the saturation power P_{sat}.

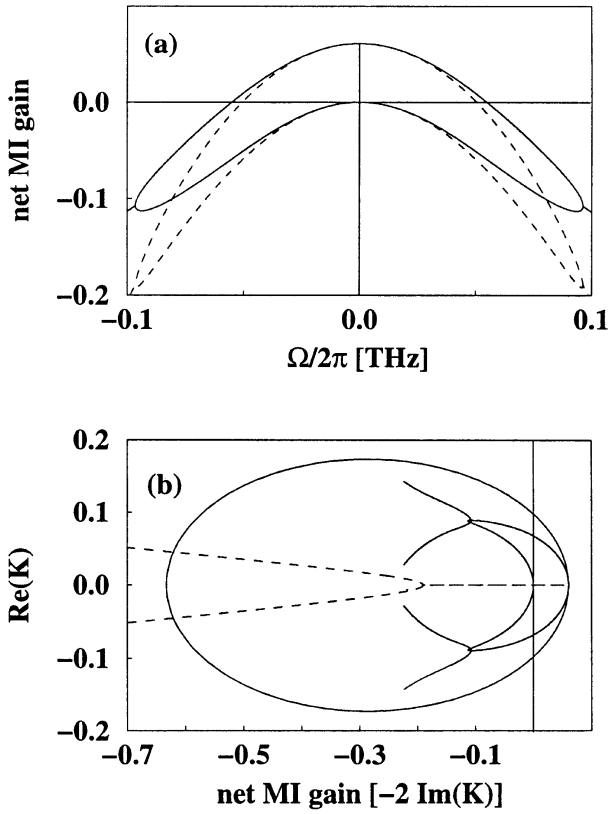


Fig. 17. Gain spectra of modulation instability (top) and trajectories of the propagation constant K (bottom) for a figure-eight laser. Solid lines show the results of the Maxwell–Bloch model, while dashed lines correspond to the Ginzburg–Landau model. Parameter values used are: $\alpha_f L = 0.4$, $g_0 L = 6$, $\beta_2 L = -0.09$ ps^2, $\theta L = 0.1$ W^{-1}, $\gamma L = 0.008$ W^{-1}, $T_2 = 1.27$ ps, $T_1 = 10^8$ ps, and $P_{\text{sat}} = 10$ mW.

In Figs. 17 and 18 the parameter θ was chosen to be positive to include the effects of an intracavity fast-saturable absorber. To study the role of pure SPM on the modulation instability, it is useful to consider the case $\theta = 0$. The interesting question is whether dopant dynamics can induce modulation instability in the normal-dispersion regime of the fiber. According to the Ginzburg–Landau model, such an instability is not possible [183, 191]. Fig. 19 shows the gain spectra for a fiber-ring laser by using $\beta_2 L = \pm 0.09$ ps^2, where L is the cavity length. In the case of anomalous dispersion, the 4 GHz wide gain spectrum is centered around 2.5 GHz, and a much narrower and much weaker peak occurs around 50 MHz. While the gain peak around 2.5 GHz vanishes in the case of normal dispersion, the narrow low-frequency gain peak survives. So, contrary to what the Ginzburg–Landau model predicts, the dopants can induce modulation instability in the normal-dispersion regime of the fiber. However, the magnitude of the gain is rather small, and it is questionable whether this instability will survive over the

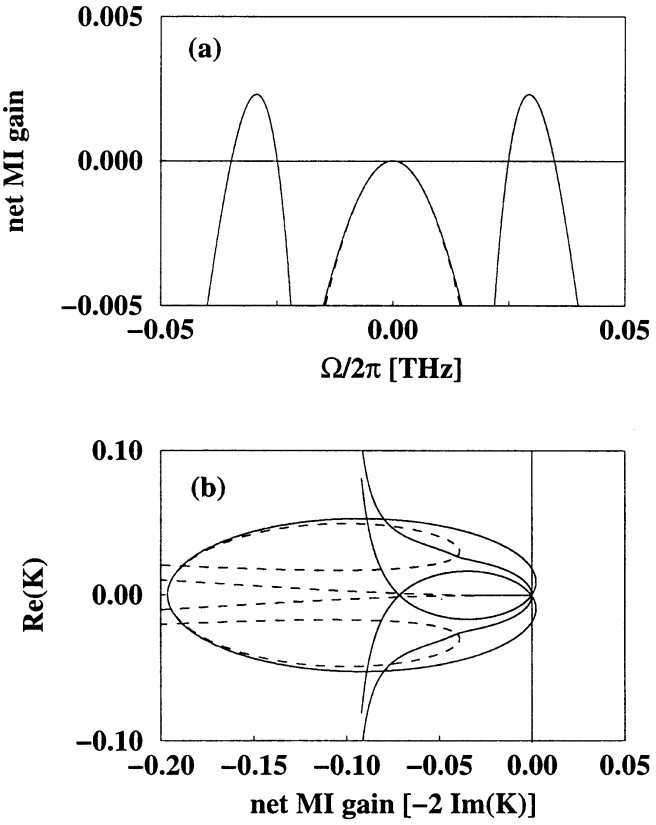


Fig. 18. Same as in Fig. 17 except for a dye laser. Parameter values used are: $\alpha_f L = 0.1$, $g_0 L = 3$, $\beta_2 L = -0.09$ ps^2, $\theta L = 0.001$ W^{-1}, $\gamma L = 0.008$ W^{-1}, $T_2 = 2.45$ ps, $T_1 = 10^3$ ps, and $P_{sat} = 1$ mW.

large number of round trips required for the growth of modulation instability. Within a few round trips, the boundary conditions of the cavity will start to have their impact on K, thus, possibly obstructing the further growth of the instability. It turns out that the detuning δ can increase the modulation-instability gain by several orders of magnitude [184].

4.3.2. Mode locking and laser instabilities

Mode locking is a well-known technique [192] that is capable of generating femtosecond optical pulses from a laser pumped continuously. It involves simultaneous operation of multiple longitudinal modes with a definite phase coherence among them. As fiber lasers have relatively long cavities ($\backsim$10 m), the longitudinal mode spacing is $\backsim$10 MHz. With a typical gain bandwidth $\backsim$5 THz, several thousands of modes can be excited simultaneously. Techniques to achieve mode locking can be divided into active and passive categories [36]. Active mode locking requires some form of external modulation at a frequency that is close to

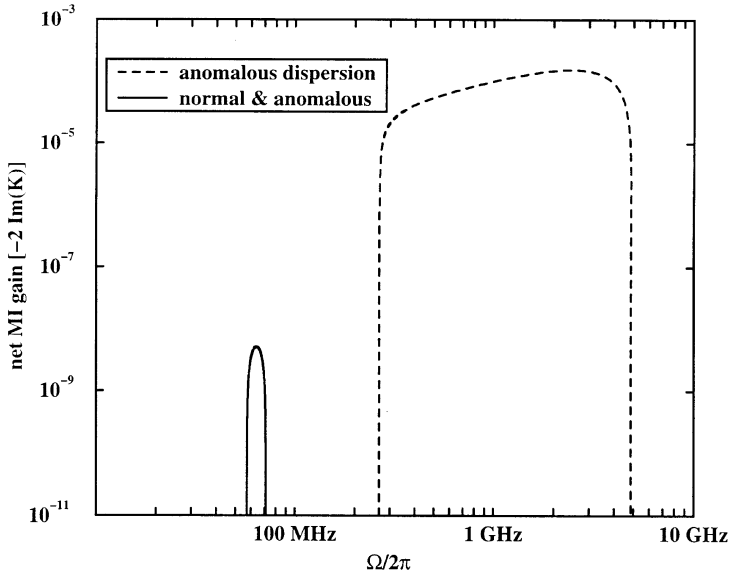


Fig. 19. Comparison of gain spectra for the cases of normal and anomalous dispersion in the absence of saturable absorption ($\theta = 0$). other parameters of the fiber laser are the same as in Fig. 17.

(or an integer multiple of) the round-trip frequency. The theory of active mode locking is well known [192].

Passive mode locking is an all-optical technique to generate short pulses from a laser. Instead of a modulator, passive mode locking uses a passive nonlinear element whose response to an optical pulse is intensity dependent. Passive mode locking can be viewed as a modulation instability that is sustained even in the presence of the feedback from cavity mirrors. Qualitatively, one can understand this statement as follows. The cavity mirrors introduce a set of longitudinal modes even in the absence of a gain medium. For a Fabry–Perot cavity of length L, the mode frequencies are $\omega_m = m\pi c/L$, and the corresponding propagation constants are given by $K_m = m\pi/L$, where m is an integer. A localized perturbation would initially grow through the convective modulation instability and fill the entire cavity after one round trip. On successive round trips, the longitudinal distribution of the perturbation is dictated by the cavity modes. Modulation instability is then only possible at the longitudinal mode frequencies ω_m. If a sufficient number of longitudinal modes experience gain through modulation instability, a pulse-like pattern will eventually emerge. After a large number of round trips, a steady state will be reached Such a steady state corresponds to the emission of a mode-locked pulse train.

Similar to the case of fiber amplifiers, the modulation-instability growth rate does not provide any information on pulses formed within the laser cavity after the onset of modulation instability. A general approach consists of solving the Maxwell–Bloch equations numerically, after including the boundary conditions at

the cavity mirrors. There are, however, quite powerful analytical models that can provide information about mode-locked pulses. These models are related to the Ginzburg–Landau equation that we introduced in Section 4.2. If we add the cavity loss and the θ term to Eq. (162) in a manner similar to Eq. (179), we obtain the following Ginzburg–Landau equation for fiber lasers:

$$\frac{\partial A}{\partial z} = \frac{1}{2}(g_0 - \alpha_c)A + \frac{1}{2}(b - i\beta_2)\frac{\partial^2 A}{\partial T^2} + (\theta + i\gamma) \mid A \mid^2 A. \tag{186}$$

By requiring that each mode-locked pulse reproduce itself after one round trip, except for a phase shift, and making the mean-field approximation so that $A(z, t) = \bar{A}(T)\exp(i\kappa z)$, where κ is a constant, the following master equation [193] is obtained from Eq. (186):

$$\left[-i\kappa + \frac{1}{2}(g_0 - \alpha_c) + \frac{1}{2}(b - i\beta_2)\frac{\partial^2}{\partial T^2} + (\theta + i\gamma) \mid \bar{A} \mid^2 \right]\bar{A} = 0, \tag{187}$$

where the GVD and SPM effects have been included through β_2 and γ, respectively. Solutions of this equation provide pulse shapes $\bar{A}(T)$ associated with mode-locked pulses.

The shape of mode-locked pulses can also be obtained by finding the solitary-wave solutions of Eq. (186). By normalizing this equation in terms of the soliton units through Eq. (155), we obtain its dimensionless form

$$i\frac{\partial u}{\partial \xi} - \frac{1}{2}(s + id)\frac{\partial^2 u}{\partial \tau^2} + \mid u \mid^2 u = \frac{i}{2}(\mu + \mu_2 \mid u \mid^2)u, \tag{188}$$

where $s = \mathrm{sgn}(\beta_2)$, and the parameters d, μ and μ_2 are defined as:

$$d = \frac{g_0 T_2^2}{\mid \beta_2 \mid}, \quad \mu = \frac{(g_0 - \alpha_c)T_0^2}{\mid \beta_2 \mid}, \quad \mu_2 = \frac{\theta \mid \beta_2 \mid}{(\gamma P_0 T_0)^2}. \tag{189}$$

Eq. (188) is non-integrable in the language of inverse scattering theory and, therefore, does not have soliton solutions in a strict mathematical sense. However, it does have solitary-wave solutions corresponding to pulses whose shape does not change with propagation. A solitary-wave solution for the case $s = -1$ (anomalous GVD) and $\mu_2 = 0$ was found as early as 1977 [194]. Differences between solitons and solitary waves are rather involved, but it suffices to say here that solitons survive collisions with each other, i.e. they retain their original shape after a collision, while solitary waves generally do not have this property. The solitary-wave solutions of Eq. (188) are sometimes called autosolitons, since all input pulses evolve toward this unique solitary wave in the case of a fiber amplifier. The presence of gain dispersion d causes the autosoliton to be chirped. The functional form for the solitary wave is [36]:

$$u(\xi, \tau) = N_s[\mathrm{sech}(p\tau)]^{1+iq}\exp(iK_s\xi), \tag{190}$$

where N_s, p, K_s and q are found by substituting Eq. (190) in Eq. (188):

$$N_s^2 = \frac{1}{2}p^2[s(q^2-2)+3qd], \tag{191}$$

$$p^2 = -\mu[s(1-q^2)+2sq]^{-1}, \tag{192}$$

$$K_s = -\frac{1}{2}p^2[s(1-q^2)-2qd], \tag{193}$$

and q is a solution of the quadratic equation

$$(d+\mu_2 s/2)q^2 - 3(s-\mu_2 d/2)q - (\mu_2 s + 2d) = 0. \tag{194}$$

The new feature of this solitary-wave solution is that it represents a chirped pulse. Moreover, such solitary waves exist in both the normal- and anomalous-GVD regime of the fiber. In the case of fiber lasers, this solution represents a mode-locked pulse whose width, chirp and amplitude depend on many laser parameters, including the cavity length [193].

Saturable absorbers play an important role in inducing mode locking. It is hard to find saturable absorbers that can respond on a time scale shorter than the pulse width when passive mode-locking is used to generate ultrashort optical pulses. A technique known as *additive-pulse mode-locking* [193, 195] converts phase changes induced by SPM to intensity changes using an interferometer, resulting in intensity-dependent transmission similar to that obtained from a fast responding saturable absorber. Using this technique, pulse widths shorter than 100 fs have been obtained from fiber lasers [36]. Recently, a figure-eight fiber laser was shown to generate 125 fs mode-locked pulses with an energy of 0.5 nJ at a very stable repetition rate [196]. For a more detailed account of passive mode-locking techniques we refer to Ref. [36].

A question that is attracting attention in recent years concerns the long-time stability of mode-locked pulse trains. In a recent study [197] the long-time stability of passively mode-locked lasers was studied by using a stability analysis based on the so-called aberrationless approximation. An analytic criterion for the stability of mode-locked pulses was found for the case of weak saturation of the absorber. In another study, long-time stability of a passively mode-locked laser was studied under pulsed injection-locking [198]. Period-doubled and even quasi-periodic types of mode-locked pulse trains have been observed experimentally [199]. Numerical solutions of a set of model equations inspired by the master-equation approach [200] agreed qualitatively with the experimental results. If one views mode locking as the consequence of the second threshold for a laser, destabilization of the mode-locked pulse train above a certain pumping level can be called the 'third laser threshold' even though this terminology is not currently used. In the next section we introduce a modal description of lasers that is capable of predicting both the second and the third thresholds.

4.3.3. Absolute instabilities in fiber lasers

The question concerning the long-term stability of a CW or mode-locked laser is clearly important. Its answer generally requires numerical solutions of the Maxwell–Bloch equations, with the appropriate boundary conditions. Since such an approach hinders physical insight, we use a model description and derive a set of multimode equations that allow us to study the long-term dynamics of a fiber laser under both CW and mode-locked operation. The linear-stability analysis of these equations governs laser instabilities that are 'absolute' in nature. Absolute modulation instability in a passive ring cavity has been studied extensively in recent years [201–203]. Lasers based on such instabilities have also been proposed and demonstrated [204–206].

Eq. (167) describes both the short- and long-term evolution of the intracavity optical field. Since the steady state of the laser is obtained by replacing $\partial A/\partial z$ with $i\kappa A$ (see Section 4.3.2.), we can interpret $T = t - z/v_g$ as the short-term propagation variable and z as the long-term spatial variable [193]. The long-term time scale can then be introduced as $t_R = z/v_g$; it measures the evolution of the field in units of the round-trip time within the laser cavity. We will treat t_R as a continuous variable, keeping in mind that it describes field evolution over multiple round trips within the laser cavity.

In general, a laser can operate in several longitudinal modes simultaneously. In most ring-cavity lasers, these modes are assumed to have a uniform intensity profile along the cavity perimeter. This assumption breaks down even for a Fabry–Perot cavity, in which the modal distribution varies on a wavelength scale (or on the scale of an optical period) because of interference between the counterpropagating waves. In a laser in which the effects of host dispersion and nonlinearities cannot be neglected, a temporal profile $f_m(T)$ can be associated with the mth longitudinal mode, where $T = t - z/v_g$. Thus, the steady-state field for a laser operating in several longitudinal modes should be written as $A_s(T) = \Sigma_m a_m f_m(T)$. If this steady state is perturbed, the expansion coefficients a_m become a function of t_R. We can study the long-term stability of the steady state by expanding the field A and the polarization B into a set of longitudinal modes as:

$$A(t_R, T) = \sum_m a_m(t_R) f_m(T), \tag{195}$$

$$B(t_R, T) = \sum_m b_m(t_R) f_m(T). \tag{196}$$

To study the long-term stability, we need to find how a_m and b_m change over multiple round trips within the laser cavity. For this purpose, we substitute Eqs. (195) and (196) into the Eqs. (167) and (168), multiply them by f_q^*, and integrate over one round-trip time T_r. Noting that the Bloch Eqs. (168) and (169) remain the same for both time scales T and t_R, we obtain:

$$\frac{1}{v_g}\sum_m \frac{da_m}{dt_R}\Gamma_{mq} = \frac{i}{2}\sum_m b_m\Gamma_{mq} - \frac{1}{2}\alpha_c\sum_m a_m\Gamma_{mq} - \frac{i}{2}\beta_2\sum_m a_m D_{mq} + (\theta + i\gamma)\sum_{m,n,p} a_m a_n^* a_p C_{mnpq}, \tag{197}$$

$$T_2\sum_m \frac{db_m}{dt_R}\Gamma_{mq} = -(1 - i\delta)\sum_m b_m\Gamma_{mq} - i\sum_m a_m g_{mq}, \tag{198}$$

where the coefficients Γ_{mq}, D_{mq} and C_{mnpq} are defined as:

$$\Gamma_{mq} = \int_0^{T_r} f_m f_q^* dT, \tag{199}$$

$$D_{mq} = \int_0^{T_r} \frac{d^2 f_m}{dT^2} f_q^* dT, \tag{200}$$

$$C_{mnpq} = \int_0^{T_r} f_m f_n^* f_p f_q^* dT, \tag{201}$$

and α_c includes the total cavity loss in a distributed sense.

The gain coefficient g_{mq} in Eq. (198) is given by:

$$g_{mq} = \int_0^{T_r} g(t_R, T) f_m(T) f_q^*(T)\, dT \tag{202}$$

and should be interpreted as follows. When $m = q$, g_{mq} represents the modal gain for the mode f_q, while for $m \neq q$, g_{mq} represents gain modulation because of beating of the longitudinal modes f_m and f_q (a phenomenon referred to as population pulsations). From Eq. (169), we readily obtain the following rate equation for the gain coefficients:

$$T_1\frac{dg_{pq}}{dt_R} = g_0\Gamma_{pq} - g_{pq} - \frac{i}{2P_{sat}}\sum_{m,n}(a_m^* b_n - a_n^* b_m)C_{mnpq}. \tag{203}$$

So far, we have not made any assumption about the orthogonal nature of longitudinal modes. The possibility that longitudinal modes may become nonorthogonal in some lasers has attracted considerable attention in recent years [192, 207, 208]. For fiber lasers, longitudinal modes are expected to be nearly orthogonal. If we assume that f_m are also normalized, the coefficients Γ_{mq} reduce to δ_{mq}, where the Kronecker $\delta_{mq} = 0$ if $m \neq q$. Eqs. (197), (198) and (203) then reduce to the following set of Lorenz–Haken-type multimode equations:

$$\frac{1}{v_g}\frac{da_q}{dt_R} = \frac{i}{2}b_q - \frac{1}{2}\alpha_c a_q - \frac{i}{2}\beta_2\sum_m a_m D_{mq} + (\theta + i\gamma)\sum_{m,n,p} a_m a_n^* a_p C_{mnpq}, \tag{204}$$

$$T_2 \frac{\mathrm{d}b_q}{\mathrm{d}t_\mathrm{R}} = -(1 - i\delta)b_q - i\sum_m a_m g_{mq}, \tag{205}$$

$$T_1 \frac{\mathrm{d}g_{pq}}{\mathrm{d}t_\mathrm{R}} = g_0\delta_{pq} - g_{pq} - \frac{i}{2P_\mathrm{sat}}\sum_{m,n}(a_m^* b_n - a_n^* b_m)C_{mnpq}. \tag{206}$$

Eqs. (204)–(206) describe the dynamics of various longitudinal modes of the laser. They include the effects of host dispersion and nonlinearity through the terms containing β_2 and γ in Eq. (204). What constitutes SPM in a propagation-based description splits into several different kinds of nonlinearities in the modal description. The triple sum in Eq. (204) describes the phenomenon of SPM when $m = n = p = q$, cross-phase modulation for $m = q$ and $n = p$, and four-wave mixing for other combinations of m, n and p [36].

In principle, Eqs. (204)–(206) can be used to study the phenomenon of passive mode locking, by including a larger and larger number of longitudinal modes as mode-locked pulses become shorter and shorter. When the mode-locked pulse pattern becomes unstable, the laser may switch to a higher harmonic mode-locked operation [38] or even enter a chaotic regime [199]. Eqs. (204)–(206) can be used to describe such behavior. When the steady-state pattern $A_\mathrm{s}(T)$ does not vary too rapidly, one can restrict the analysis to a small number of modes, although such a restriction will exclude passive mode-locking.

4.3.4. Single-mode absolute instabilities

As an example of the usefulness of the modal description, we consider how the GVD and SPM effects modify the second threshold of a fiber laser. The simplest case, analogous to the single-mode Lorenz–Haken model, is obtained from Eqs. (204)–(206) by setting $m = n = p = q = 1$. The resulting single-mode equations are:

$$\frac{1}{v_\mathrm{g}}\frac{\mathrm{d}a_1}{\mathrm{d}t_\mathrm{R}} = \frac{i}{2}b_1 - \frac{\alpha_\mathrm{c}}{2}a_1 + (\theta + i\gamma)C_{1111} \mid a_1 \mid^2 a_1, \tag{207}$$

$$T_2 \frac{\mathrm{d}b_1}{\mathrm{d}t_\mathrm{R}} = -(1 - i\hat{\delta})b_1 - ia_1 g_{11}, \tag{208}$$

$$T_1 \frac{\mathrm{d}g_{11}}{\mathrm{d}t_\mathrm{R}} = g_0 - g_{11} + \frac{\mathrm{Im}(a_1^* b_1)}{P_\mathrm{sat}}. \tag{209}$$

The effect of GVD on a single longitudinal mode merely involves a frequency shift of magnitude $\omega_\mathrm{D} = \beta_2 v_\mathrm{g} D_{11}/2$, which has been scaled out by changing the carrier frequency ω_0 to $\omega_0 + \omega_\mathrm{D}$. This is equivalent to resetting the atomic detuning δ to $\hat{\delta} = \delta + \omega_\mathrm{D}$. We normalize these equations using the standard Lorenz–Haken notation introduced in Section 2 and obtain:

$$\frac{dx}{dt} = -\sigma(x - y) + iqx \mid x \mid^2, \tag{210}$$

$$\frac{dy}{dt} = -(1 - i\hat{\delta})y + (r - z)x, \tag{211}$$

$$\frac{dz}{dt} = -bz + \mathrm{Re}(x^* y), \tag{212}$$

where the normalized time $t = t_R/T_2$ has been introduced. The parameters σ, b, r and q, and the variables x, y and z, are related to quantities appearing in Eqs. (207)–(209) as:

$$\sigma = \alpha_c v_g T_2/2, \quad b = T_2/T_1, \quad r = g_0/\alpha_c, \quad q = (\gamma + i\theta) C_{1111} P_{sat} T_1 v_g, \tag{213}$$

$$x = a_1 (b/P_{sat})^{1/2}, \quad y = (ib_1/\alpha_c)(b/P_{sat})^{1/2}, \quad z = r - g_{11}/\alpha_c. \tag{214}$$

These equations are identical to the standard Lorenz–Haken equations, Eqs. (44)–(46) of Section 2, except for the last term in Eq. (210) that is responsible for SPM and intensity-dependent absorption (IDA) occurring because of the host nonlinearities [5]. They provide the framework for studying the effects of SPM and IDA on the second threshold of a laser.

We follow the same method as that used in Section 2 to find the second laser threshold and study two cases in detail. One is characterized by the combination $\sigma = 3$ and $b = 1$ and is called the 'bad-cavity' laser, while the other ('good-cavity' laser) has the combination $\sigma = b = 1$. See Section 2 for a detailed description of the stability properties of these two laser systems. To separate the effects of SPM and IDA, we introduce the real and imaginary parts of the parameter q through $q = q' + q''$ and treat q' and q'' as two independent parameters.

We first consider a laser without IDA and set $q'' = 0$. SPM is then the only host-induced nonlinearity. In Fig. 20 we show the pump parameter $r_{th}^{(2)}$ at the second threshold as a function of the SPM parameter q' for the bad-cavity laser operating on-resonance ($\delta = 0$). In absence of SPM ($q' = 0$), the second threshold occurs at $r = 21$. Quite surprisingly, the global effect of SPM is to stabilize the laser: already at $|\gamma'| = 0.04$ the second threshold ceases to exist. This result is surprising since SPM is essential for convective instabilities to occur in passive fibers and fiber amplifiers. From the steady-state analysis we know that SPM causes a shift in operating frequency away from the gain peak, thus lowering the output power. From a stability point of view, this shift detunes the laser and increases its second threshold. The good-cavity laser ($\sigma = b = 1$) shows no second threshold with or without SPM.

Quite a different picture emerges when IDA is included. For both good- and bad-cavity lasers, IDA can reduce the second threshold dramatically, as shown in Fig. 21. In both cases, positive values of q'' account for two-photon absorption while negative values of q'' can be used to describe the effect of a fast saturable absorber. Clearly, fast saturable absorption (negative q'') reduces the second laser

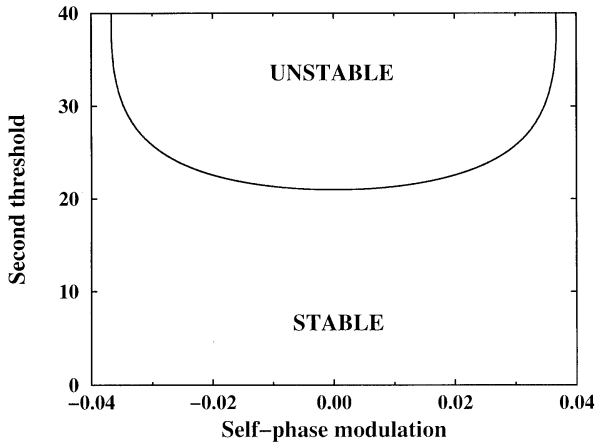


Fig. 20. Effect of self-phase modulation (SPM parameter q') on the second threshold $r_{th}^{(2)}$ of a bad-cavity laser with $\sigma = 3$ and $b = 1$.

threshold for both good- and bad-cavity lasers. In the presence of a fast saturable absorber, self-pulsing can, therefore, occur at pump levels only two times the first threshold. We stress that this self-pulsing is not related to passive mode locking because of our assumption of a single longitudinal mode. In fact, the frequency of self-pulsing is related to relaxation oscillations and is a fraction of the longitudinal-mode spacing.

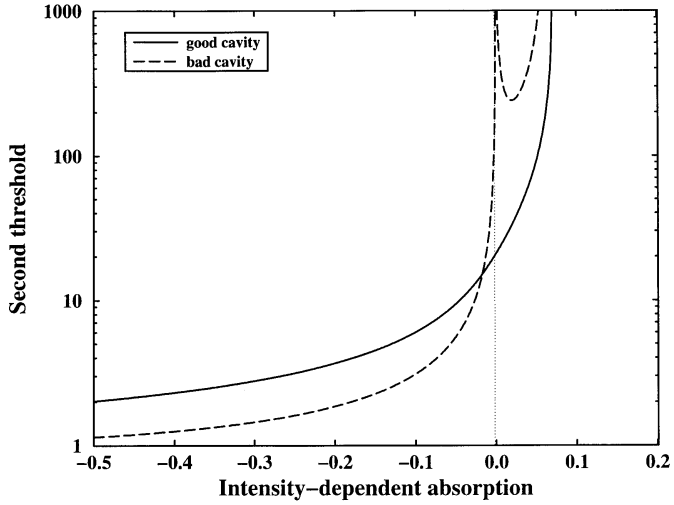


Fig. 21. Effect of intensity-dependent absorption (IDA parameter q'') on the second threshold $r_{th}^{(2)}$. Dashed line: a good-cavity laser with $\sigma = 1$ and $b = 1$. Solid line: a bad-cavity laser with $\sigma = 3$ and $b = 1$.

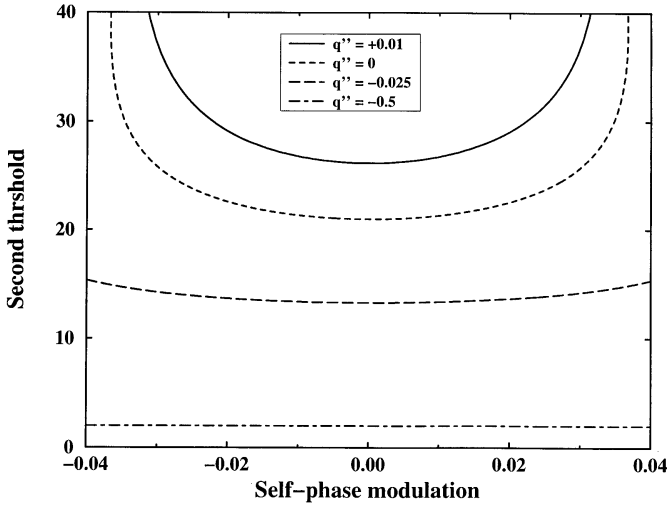


Fig. 22. Second threshold $r_{\mathrm{th}}^{(2)}$ for a bad-cavity laser ($\sigma = 3$ and $b = 1$) plotted as a function of the SPM parameter $q\prime$ for four different values of $q\prime$.

Fig. 22 shows the combined effect of SPM and IDA for the bad-cavity laser. The second laser threshold is shown as a function of the SPM parameter $q\prime$ for four different values of q''. The effects of SPM and negative IDA have opposite effects on the second threshold: SPM stabilizes the laser by increasing the second threshold, while negative IDA destabilizes the laser and reduces the pumping level at which the second threshold occurs. For the good-cavity laser a qualitatively similar figure is obtained, although in that case no second threshold exists when $q'' = 0$.

Fiber-ring lasers usually oscillate in many longitudinal modes simultaneously, because their mode spacing ($\backsim$10 MHz) is a fraction of the gain bandwidth (>1 THz). They can be forced to operate in a single longitudinal mode by using a wavelength-selective filter such as a fiber Bragg grating . The question that we can answer using the modal analysis is whether such a CW fiber laser would start self-pulsing above a certain pumping level. When we use typical values for a neodymium-doped fiber laser, $\alpha_c = 46\ \mathrm{km}^{-1}$, $T_2 = 1.75$ ps, $T_1 = 0.1$ ms, $\gamma = 2.3\ \mathrm{W}^{-1}\ \mathrm{km}^{-1}$, and $P_{\mathrm{sat}} = 11.4\ \mu\mathrm{W}$, the dimensionless parameters are $\sigma = 8.84 \times 10^{-6}$, $b = 1.77 \times 10^{-8}$, and $q\prime = 5.23 \times 10^{-4}$. These values make a Nd-doped fiber-ring laser a good-cavity laser with no second threshold. It turns out that even relatively small values of q'' cause the second threshold to exist at quite low pump values. For example, when $q'' = -3.65 \times 10^{-8}$, the second laser threshold exists at a pump level only twice the first laser threshold. This value of q'' corresponds to $\theta = 1.6 \times 10^{-4}\ \mathrm{W}^{-1}\ \mathrm{km}^{-1}$ if we use $C_{1111} = 1$. This is a relatively weak saturable absorption since it reduces losses in the cavity by about 0.1% for a 10 m long fiber laser with an intracavity power of 1 W. At the second

threshold, the relaxation-oscillation frequency is 50 kHz, and self-pulsing will occur at a repetition rate close to that value.

We should mention that CW fiber lasers often exhibit self-pulsing without introducing any saturable absorption intentionally. The theory of this section rules out SPM as the origin of self-pulsing in single-mode fiber lasers. However, SPM may induce self-pulsing if more than one longitudinal mode is included in the analysis. Several other explanations for the observed self-pulsing have been put forward. It was shown theoretically and experimentally that the existence of ion clusters in heavily Er-doped fiber lasers leads to self-pulsing behavior, and the same model is also applicable to dual-wavelength or bi-polarized lasers [209, 210]. Another explanation of the self-pulsing behavior of Nd-doped fiber lasers is related to the dynamics of the two polarization components, and involves the birefringence of the fiber [211, 212]. Erbium-doped fiber lasers have also been found to exhibit self-pulsing, chaos and antiphase dynamics between the different polarization components of the optical field [213]. The polarization dynamics in an erbium-doped unidirectional ring-cavity laser has recently been investigated both theoretically and experimentally [214, 215]. The theoretical analysis is based on a difference-differential equation, similar to that used for the description of the Ikeda instability in a passive ring cavity [216]. It is capable of describing absolute instabilities, but does not include the effects of GVD and SPM. Numerical simulations show good agreement with the experiments [215]. As mentioned earlier, this review does not cover polarization instabilities in detail, and we refer to the literature for further details.

5. Summary and concluding remarks

The objective of this paper was to provide a review of laser instabilities from a modern perspective. Since semiconductor and fiber lasers have matured only over the last 10–15 years, we focused heavily on them, while introducing new developments relevant to the field of laser instabilities. We have chosen not to include spatial and polarization instabilities in this review and only make a few remarks at relevant places.

In Section 2 we show how the well-known Lorenz–Haken model of laser instabilities is obtained from the Maxwell–Bloch equations, which describe the dynamics of a two-level system interacting with light. All relevant assumptions are discussed, particularly the mean-field approximation. We use the Lorenz–Haken model to introduce the basic concepts such as steady-state solutions, linear-stability analysis, first and second laser thresholds, Hopf bifurcation and self-pulsing, strange attractor in phase space, and various routes to chaos.

Section 3 is devoted to semiconductor lasers. We start with the semiconductor Bloch equations, which are considerably more involved than those obtained in Section 2 for a homogeneously broadened two-level atomic system. We discuss the necessity of dealing with them under some conditions. We give an overview of the

recent research efforts directed toward deriving a set of generalized Lorenz–Haken equations from the semiconductor Bloch equations. We focus on one model and show that a stand-alone semiconductor laser can exhibit a second threshold even when the bad-cavity condition is not satisfied. We then adiabatically eliminate the polarization dynamics by restricting our analysis to single-mode semiconductor lasers. Such lasers belong to the class-B lasers and are normally stable under CW operation, since perturbations from the steady state decay through relaxation oscillations. However, single-mode semiconductor lasers show a wealth of instabilities whenever they are exposed to external optical perturbations. We discuss the instabilities associated with injection-locking, optical feedback, and phase-conjugate optical feedback in some detail.

The description of optical instabilities in fiber amplifiers and lasers requires quite a different approach. The key issue here is that the passive fiber itself causes a convective instability, known as the modulation instability. This instability is the result of a four-wave-mixing process, phase-matched by the phenomenon of self-phase modulation. The existence of optical solitons is intimately connected with this instability. We discuss the effects of the dopant-induced gain on the modulation instability by first using a complex Ginzburg–Landau model that approximates the gain spectrum of a fiber amplifier with a parabola. We then introduce the Maxwell–Bloch equations that include the effects of fiber dispersion and nonlinearity, and discuss modulation instability by using them. Considerable differences are found between the two cases.

In a fiber laser, the cavity provides a feedback mechanism and supports only frequencies corresponding to those of the longitudinal modes. The modulation instability can survive in this environment if a sufficient number of longitudinal modes become unstable simultaneously. We discuss the relation of such an instability to passive mode locking. We show how convective modulation instability transforms into an absolute instability over multiple round trips. By using a modal decomposition, we obtain a set of multimode equations whose linear-stability analysis can predict the onset of absolute instabilities. We apply this formalism to a single-mode fiber laser and show that self-phase modulation actually leads to long-term stability of a laser, while intensity-dependent absorption causes destabilization. Fiber lasers with a nonlinear element acting as a fast-responding saturable absorber are found to have a second threshold at pumping levels only twice above the first threshold, even when cavity losses are so low that the standard Lorenz–Haken model would predict no second threshold.

In conclusion, the field of laser instabilities is far from being considered ‘dead’. Recent developments in the field of semiconductor and fiber lasers have added several new features that are not yet fully understood. In our opinion, future developments in laser physics and technology will continue to provide new impetus to the field of laser instabilities.

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